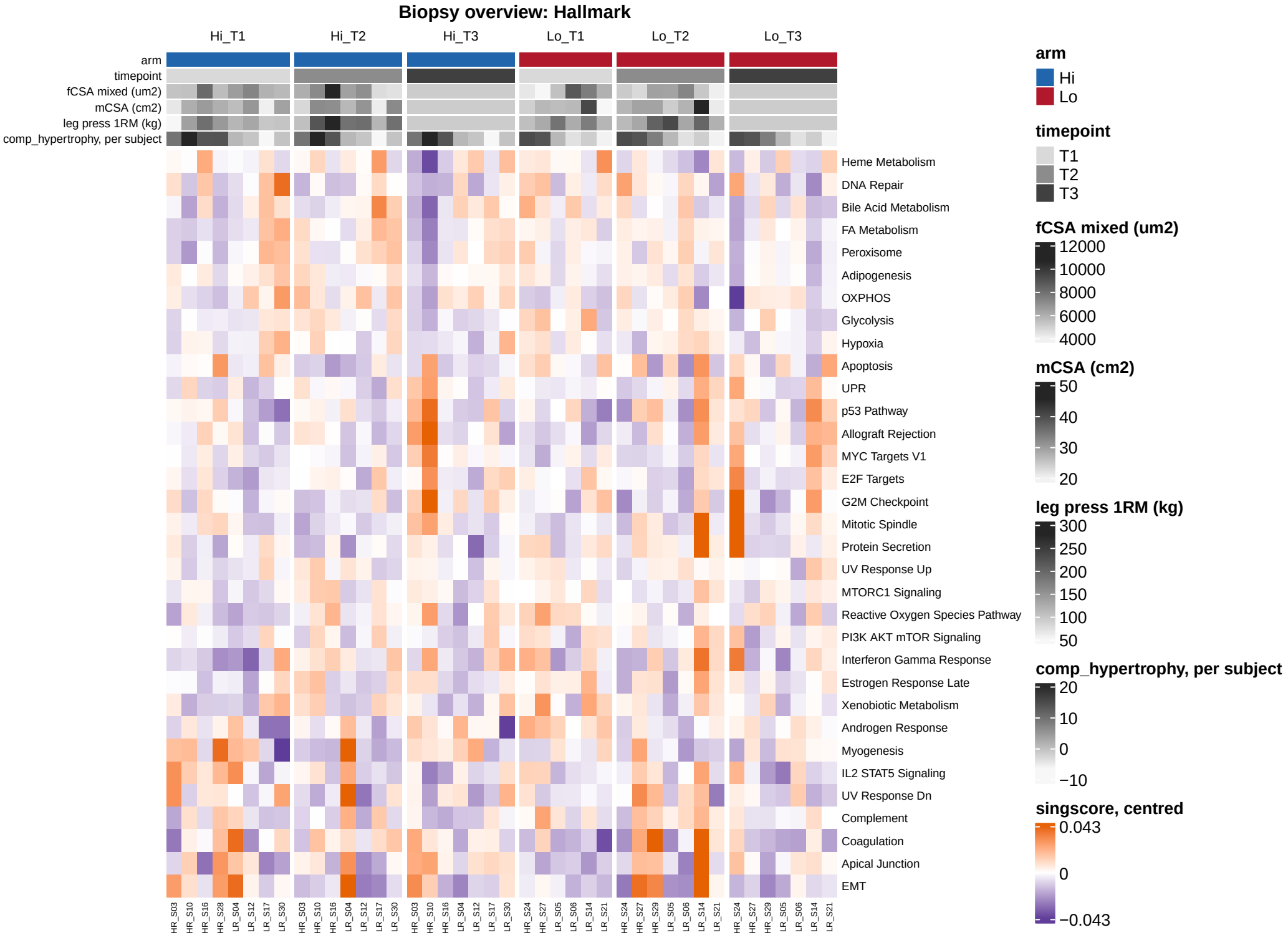
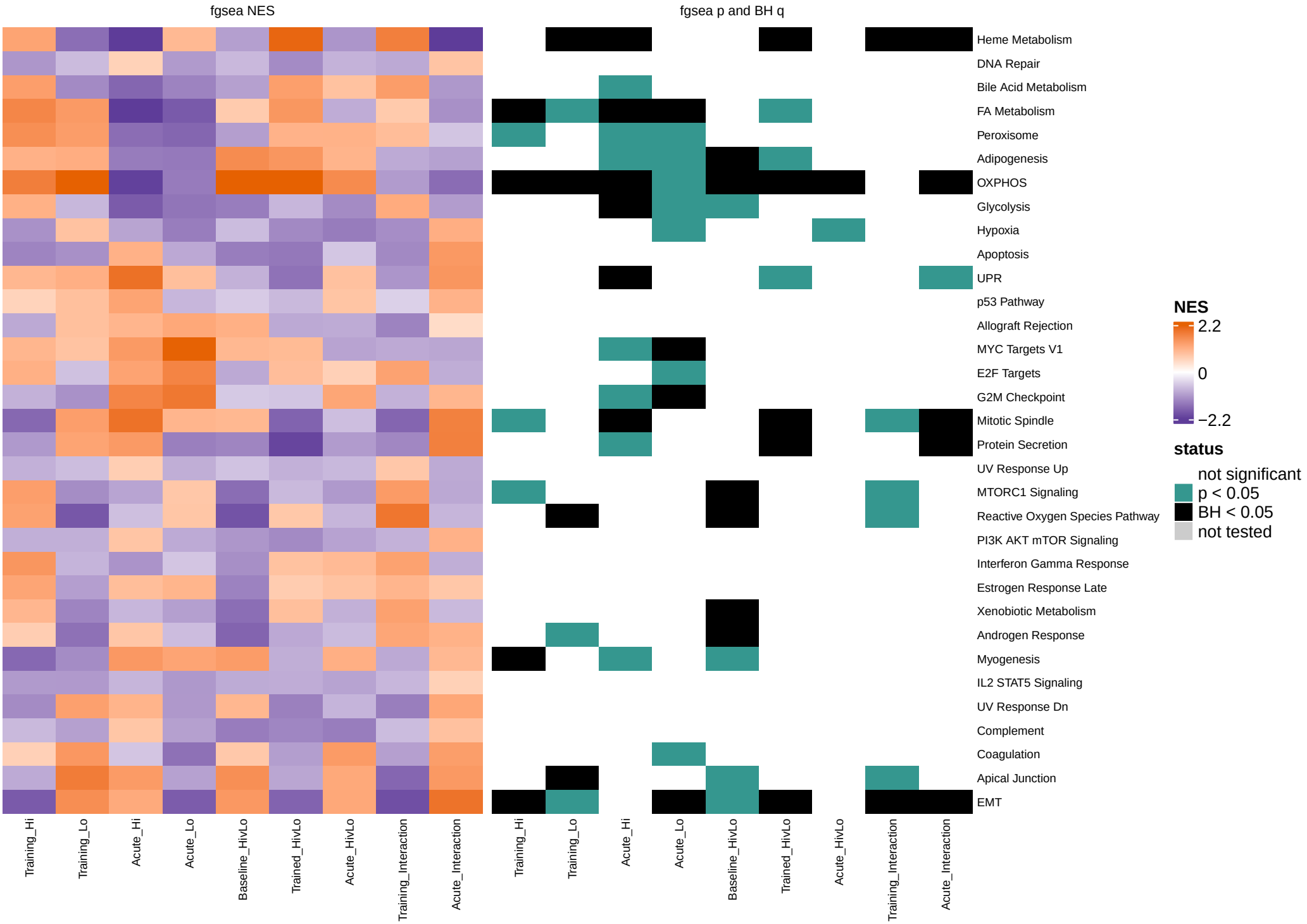
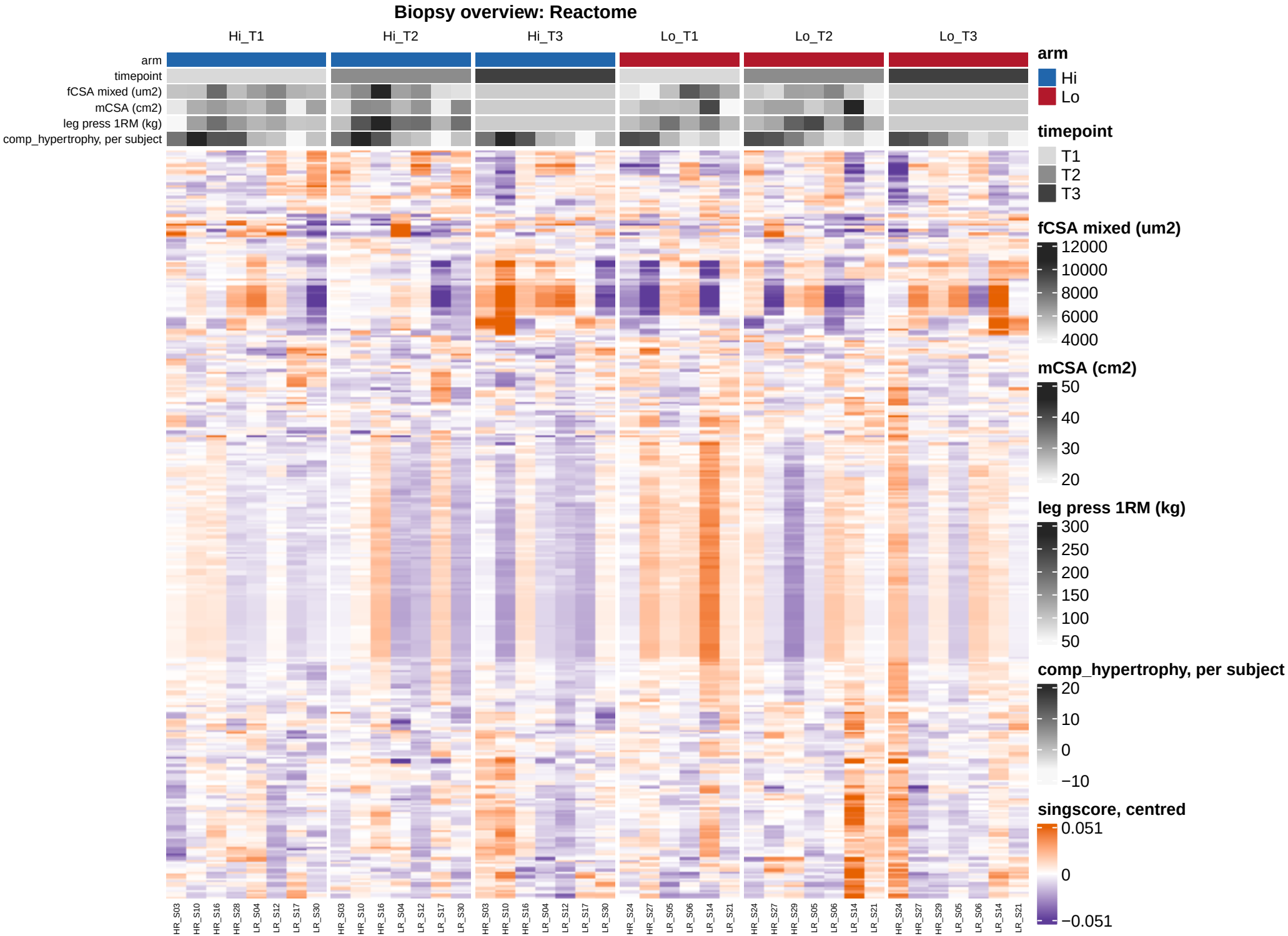


All 33 sets by 42 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.043. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: Hallmark

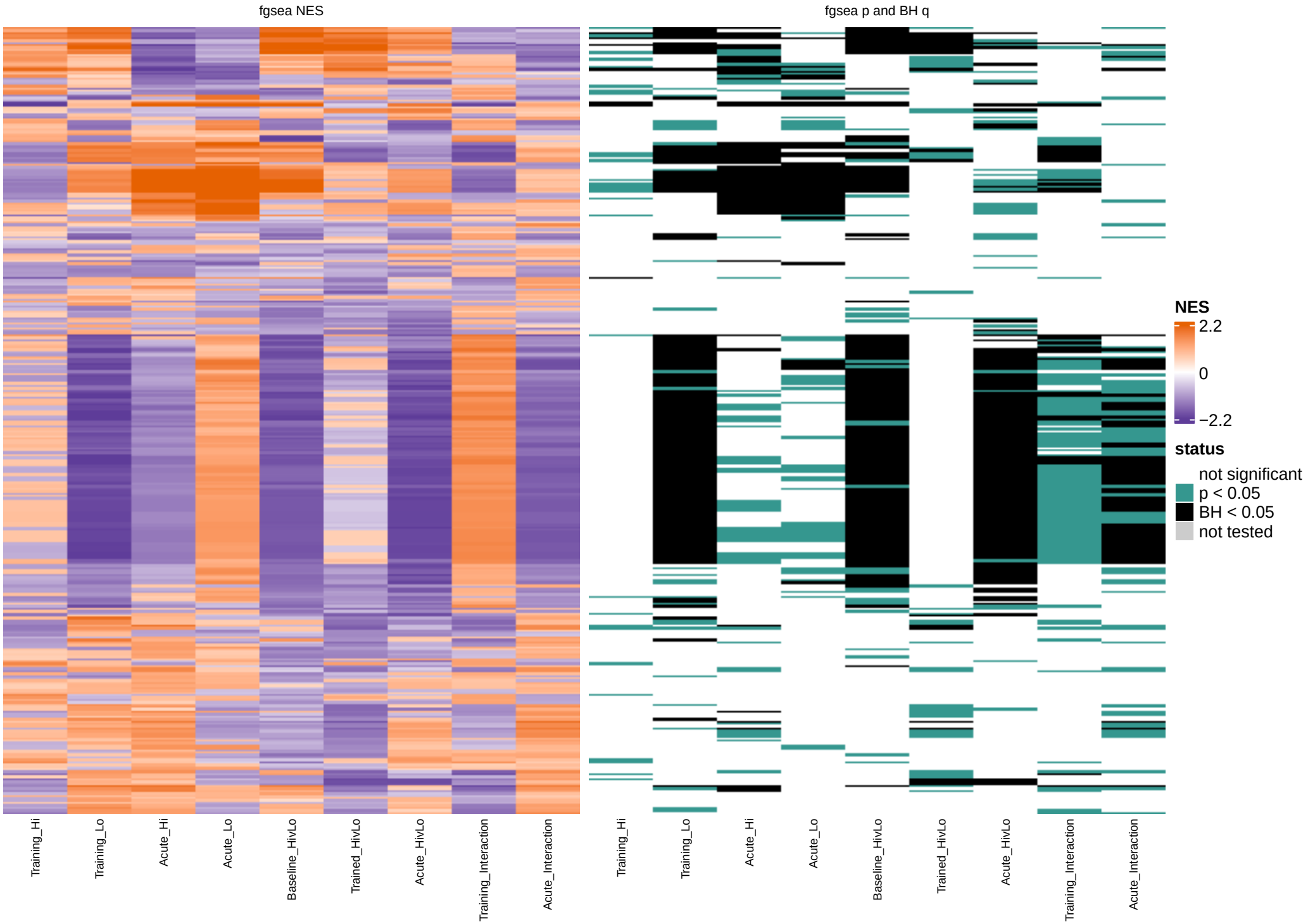


All 33 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.2. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

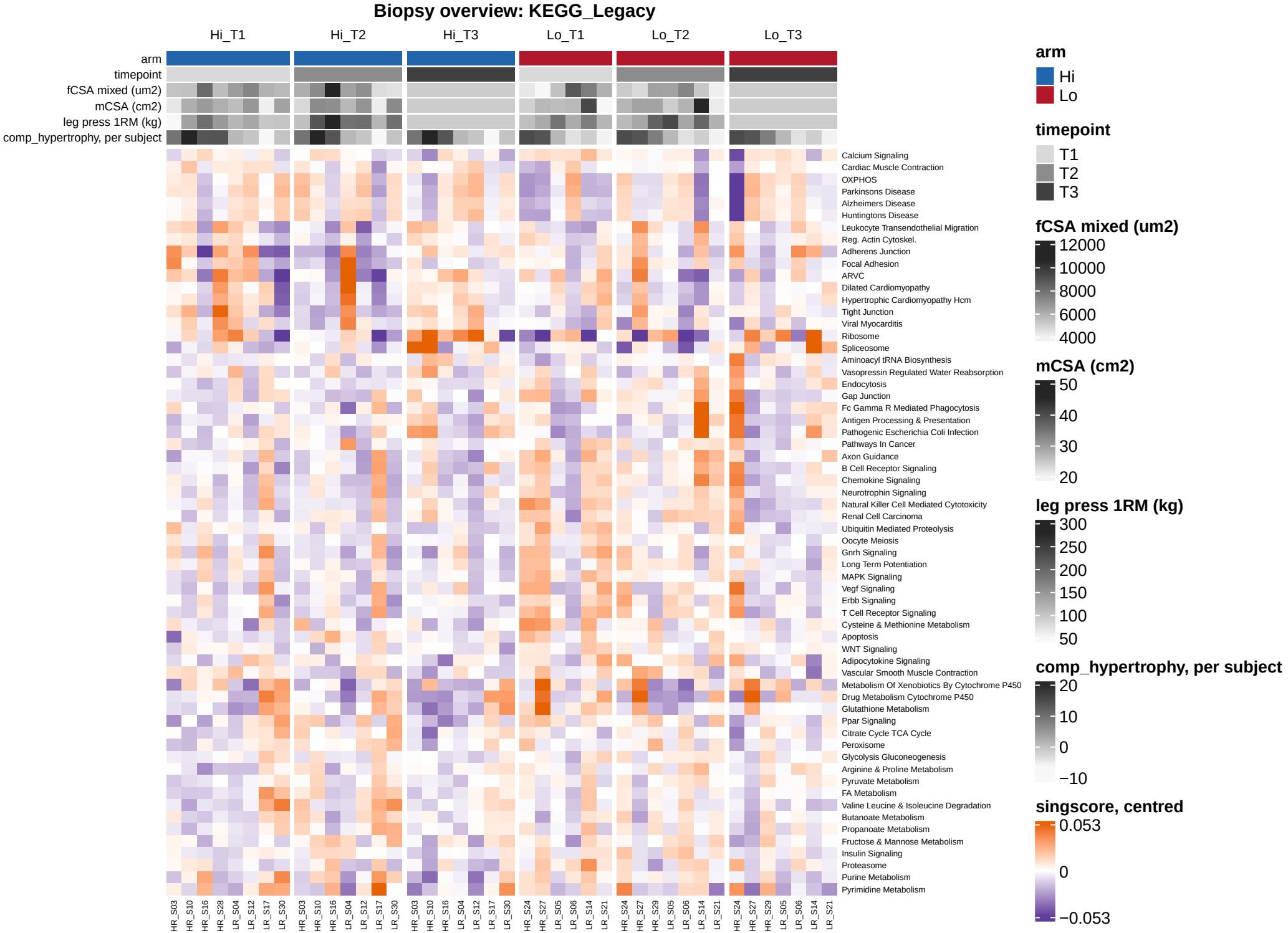


All 328 sets by 42 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.051. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: Reactome

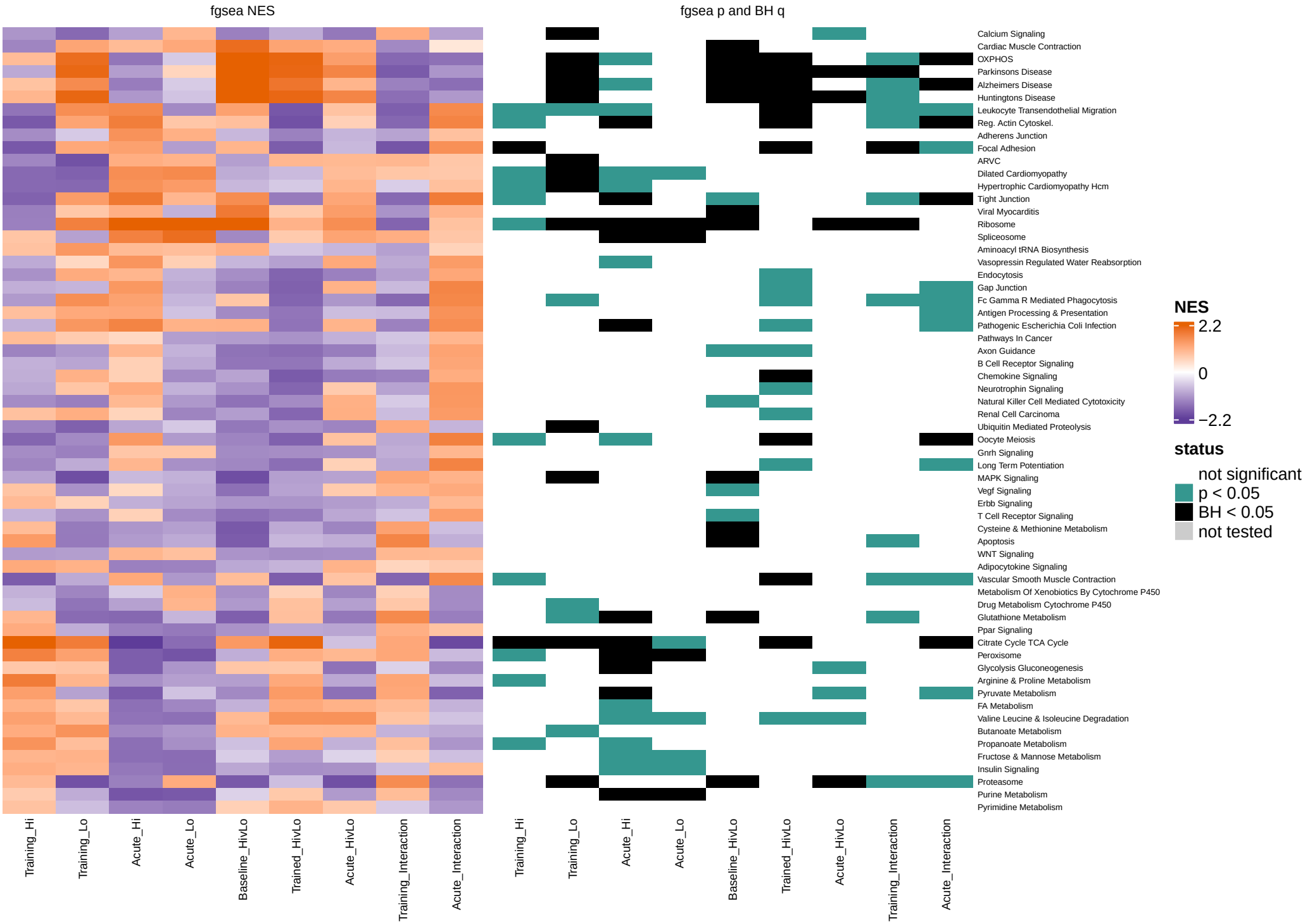


All 328 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.2. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

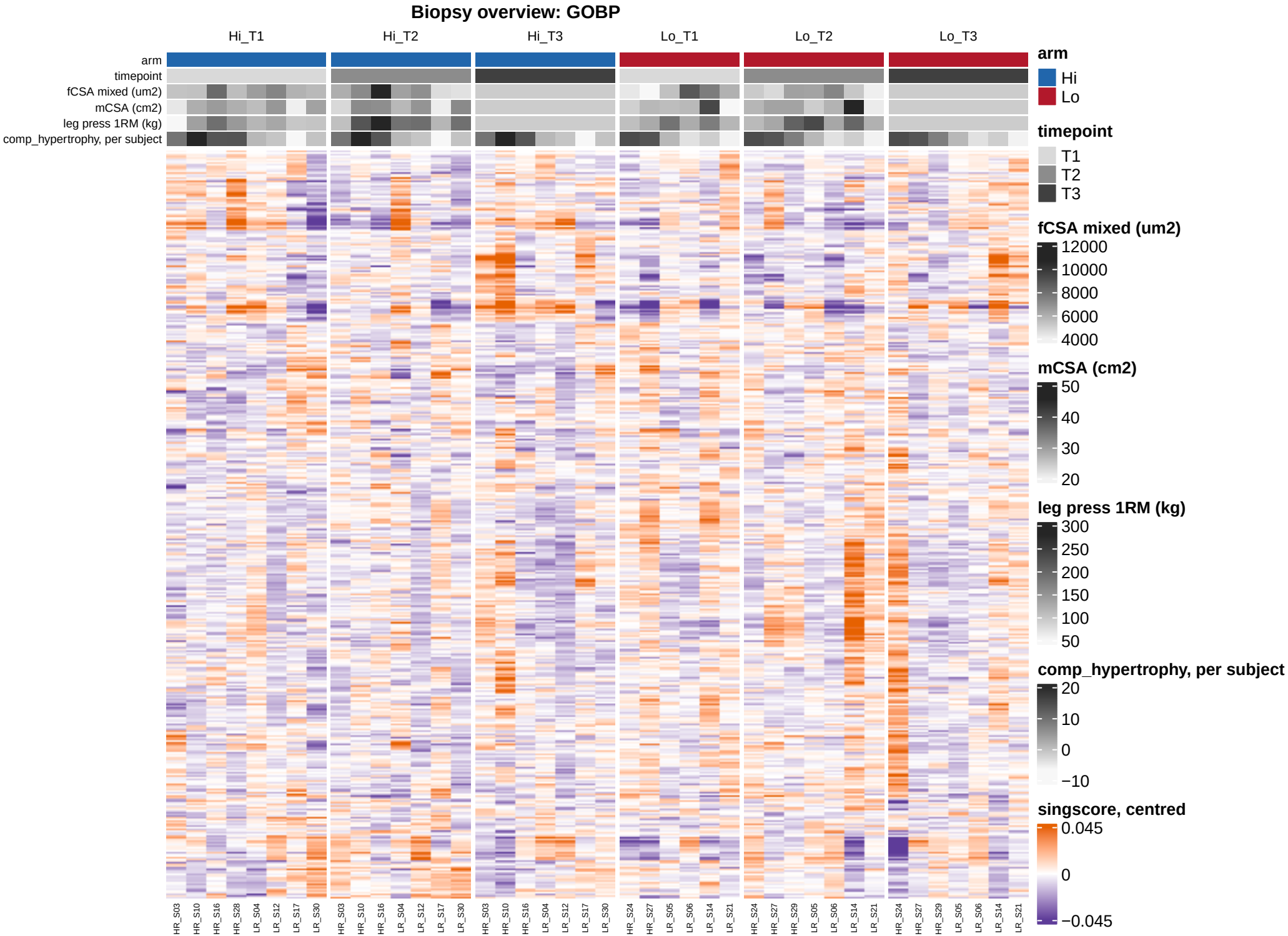


All 62 sets by 42 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.053. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: KEGG_Legacy

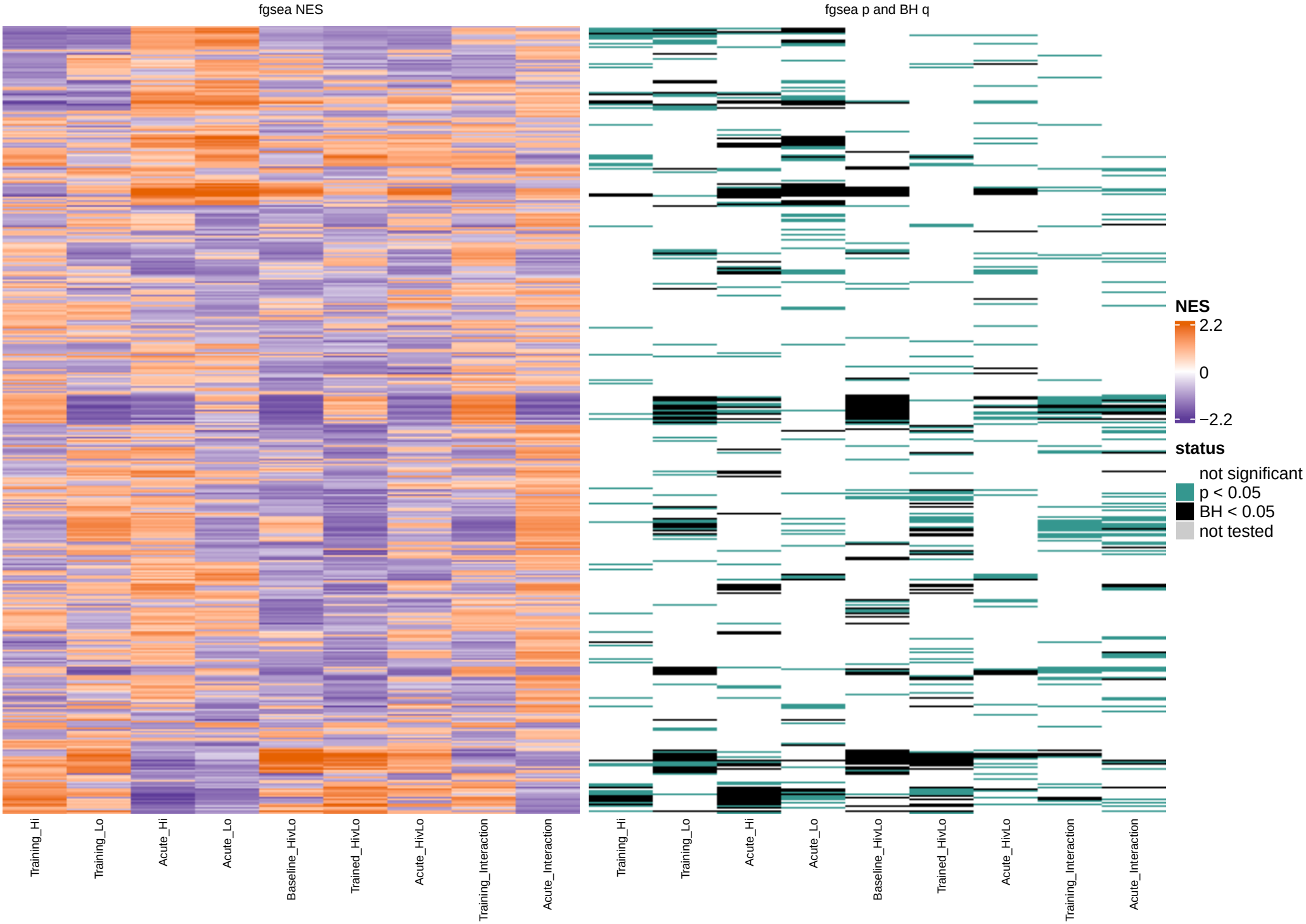


All 62 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.2. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

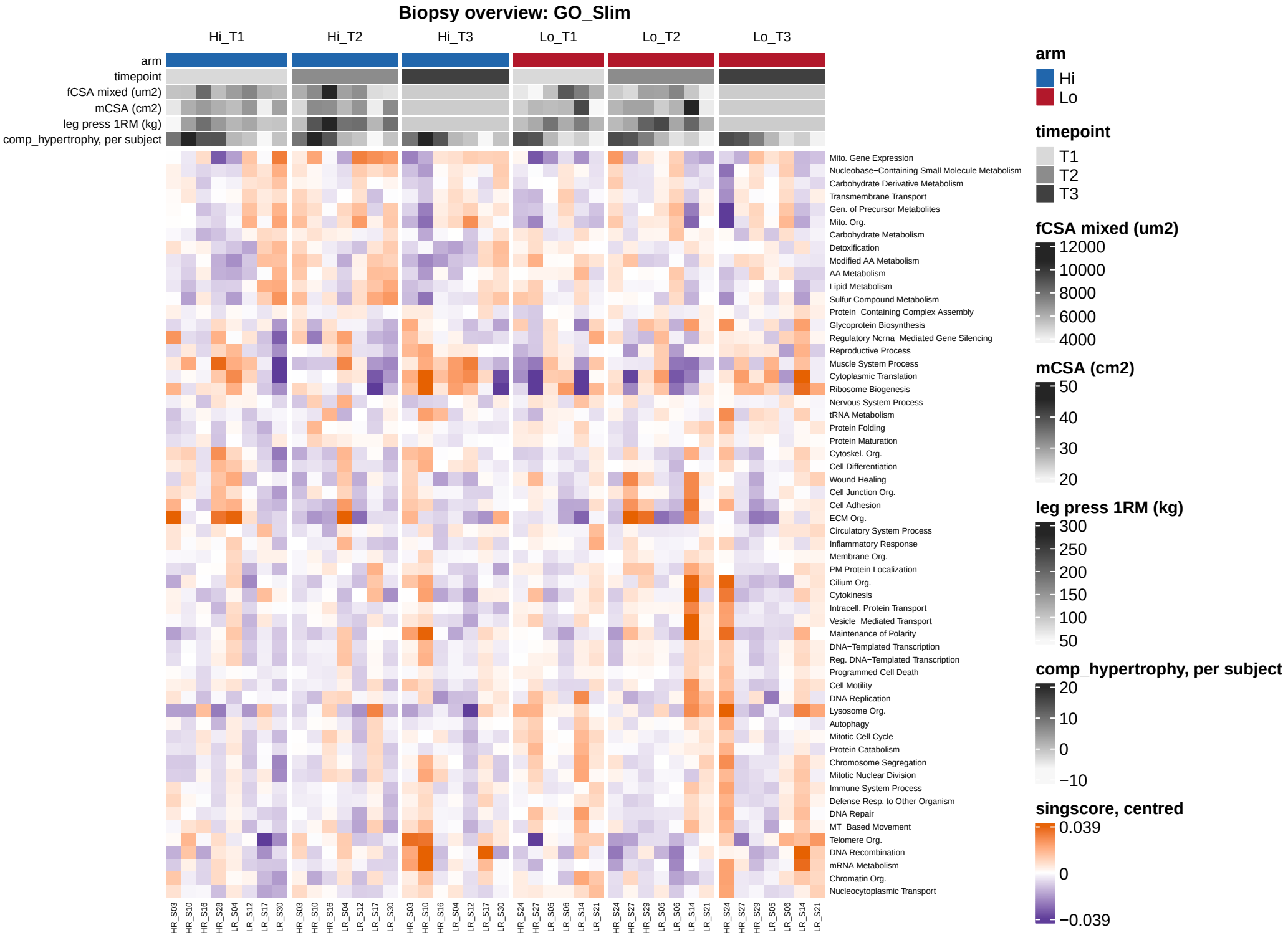


All 897 sets by 42 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.045. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

Contrast overview: GOBP

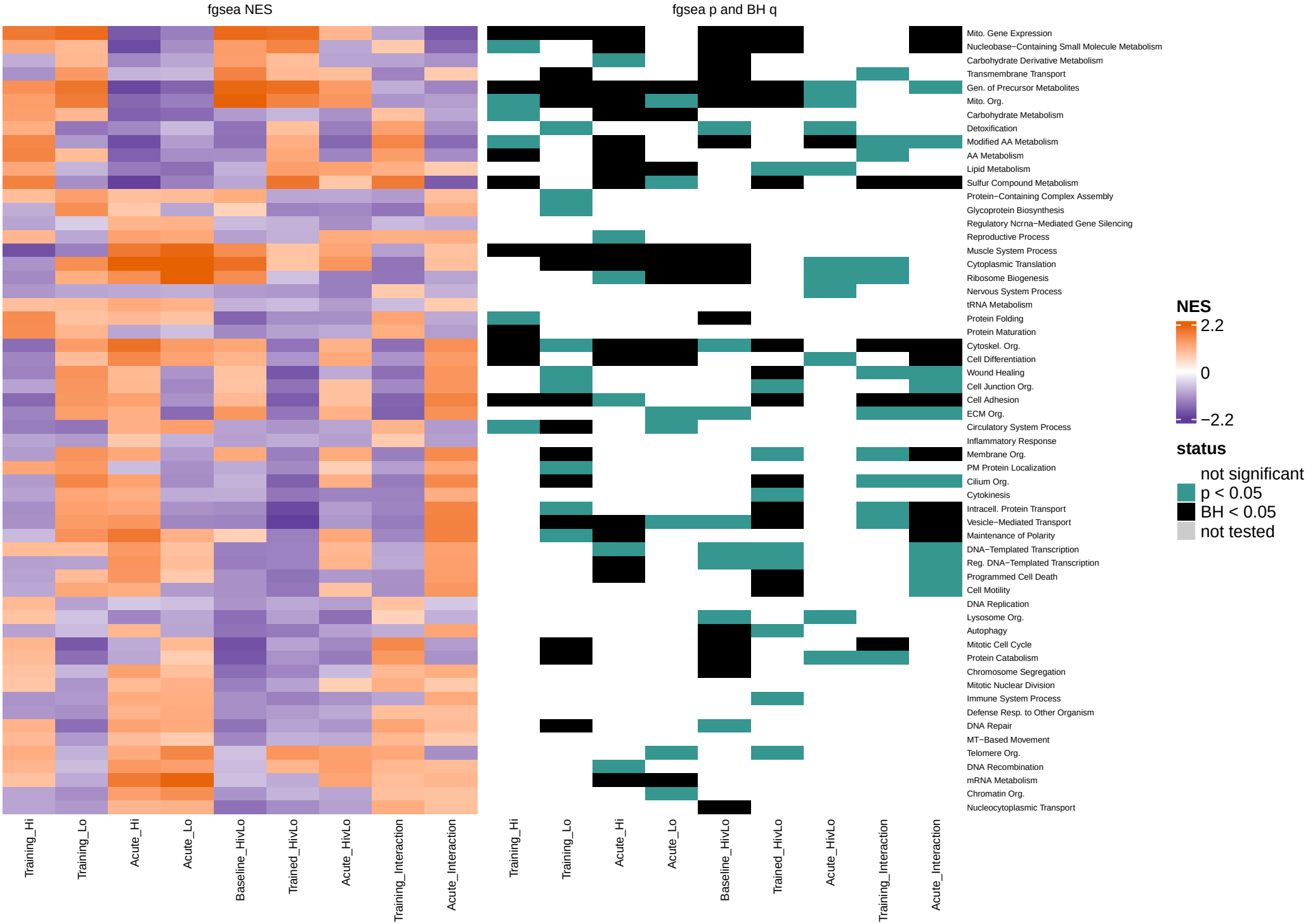


All 897 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.2. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.

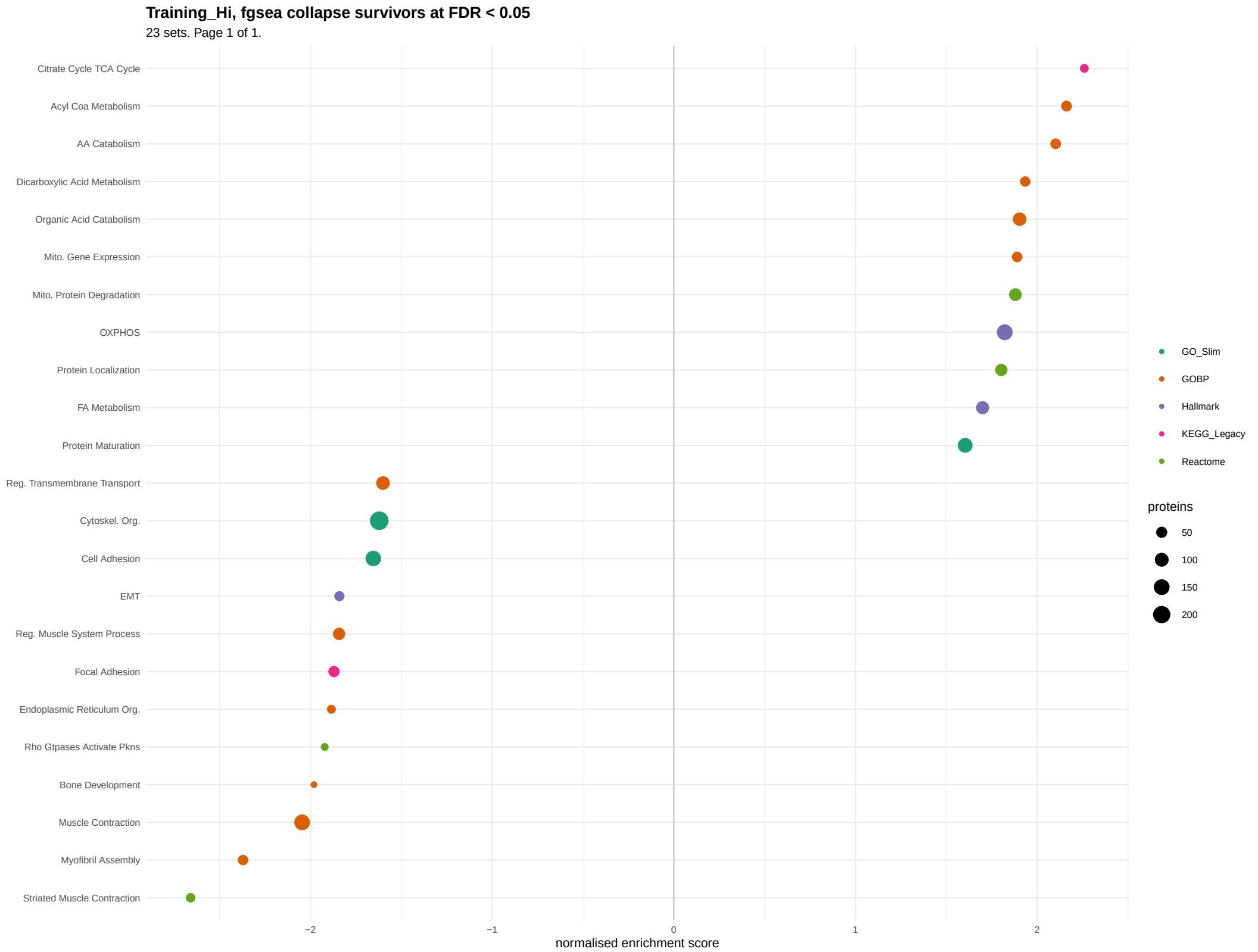


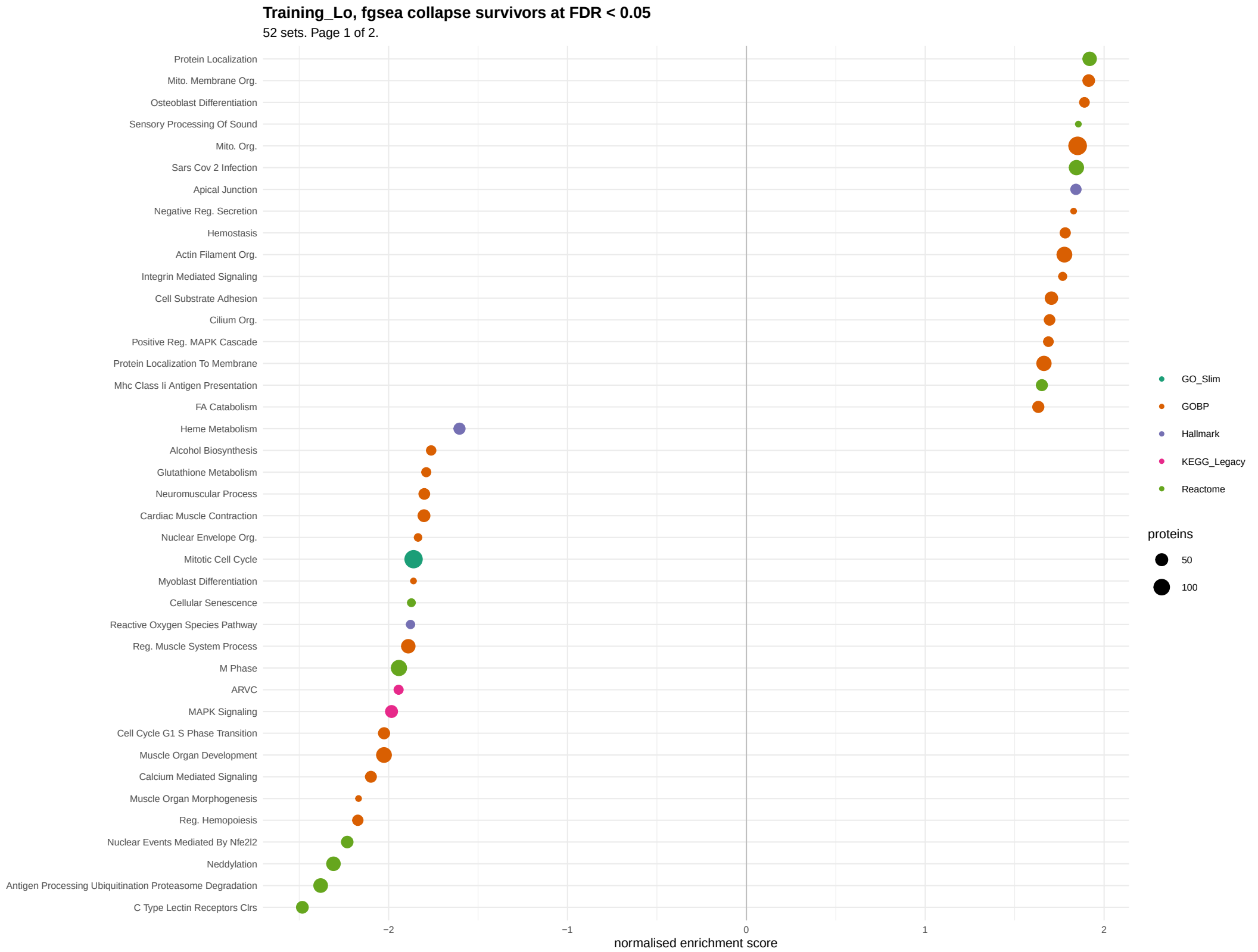
All 58 sets by 42 biopsies, grouped by arm and time point; column labels are subjects. Fill: singscore minus the set's mean over biopsies. Colours stop at the 99th percentile of |value|, 0.039. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: missing.

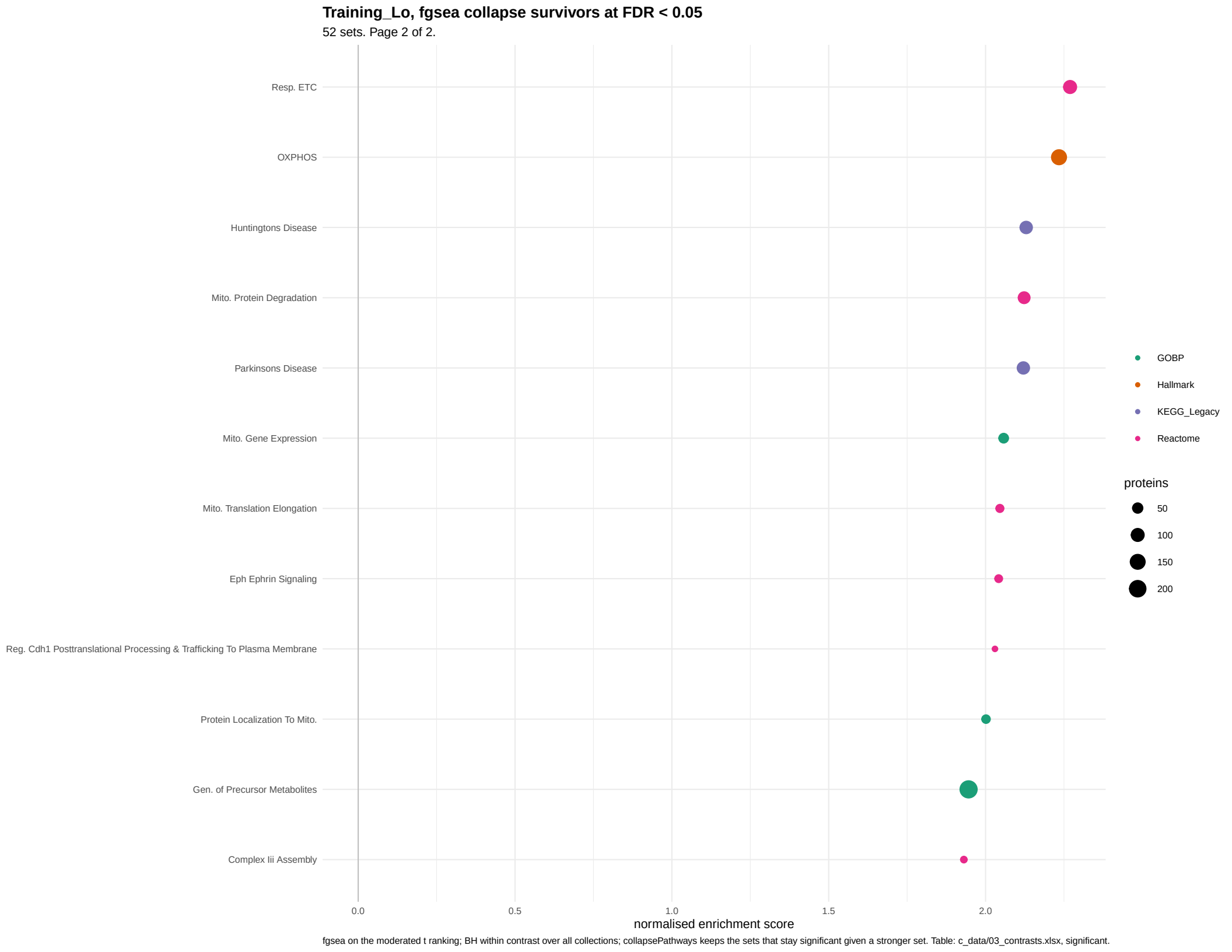
Contrast overview: GO_Slim

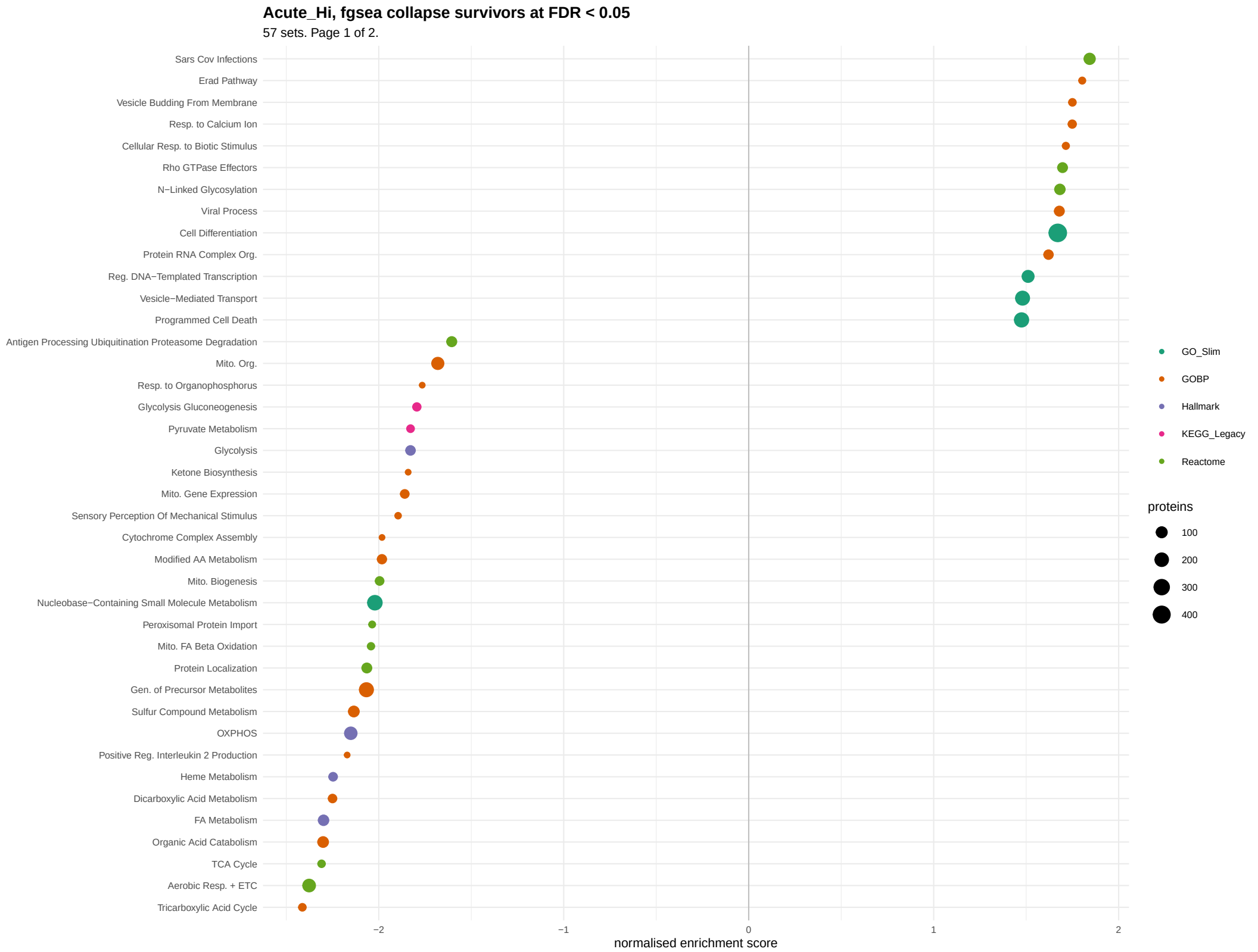


All 58 sets by the nine contrasts. Fill: fgsea normalised enrichment score; status: p and BH q within contrast, collections pooled. Colours stop at the 99th percentile of |value|, 2.2. Rows are ordered within collection by average−linkage clustering (1 − Pearson r) of singscore, for display only. Grey: not tested. Table: c_data/set_tests.rds, set_tests.


Training_Hi, fgsea collapse survivors at FDR < 0.05
23 sets. Page 1 of 1.
Citrate Cycle TCA Cycle
Acyl Coa Metabolism
AA Catabolism
Dicarboxylic Acid Metabolism
Organic Acid Catabolism
Mito. Gene Expression
Mito. Protein Degradation
OXPHOS
Protein Localization
FA Metabolism
Protein Maturation
Reg. Transmembrane Transport
Cytoskel. Org.
Cell Adhesion
EMT
Reg. Muscle System Process
Focal Adhesion
Endoplasmic Reticulum Org.
Rho Gtpases Activate Pkns
Bone Development
Muscle Contraction
Myofibril Assembly
Striated Muscle Contraction
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

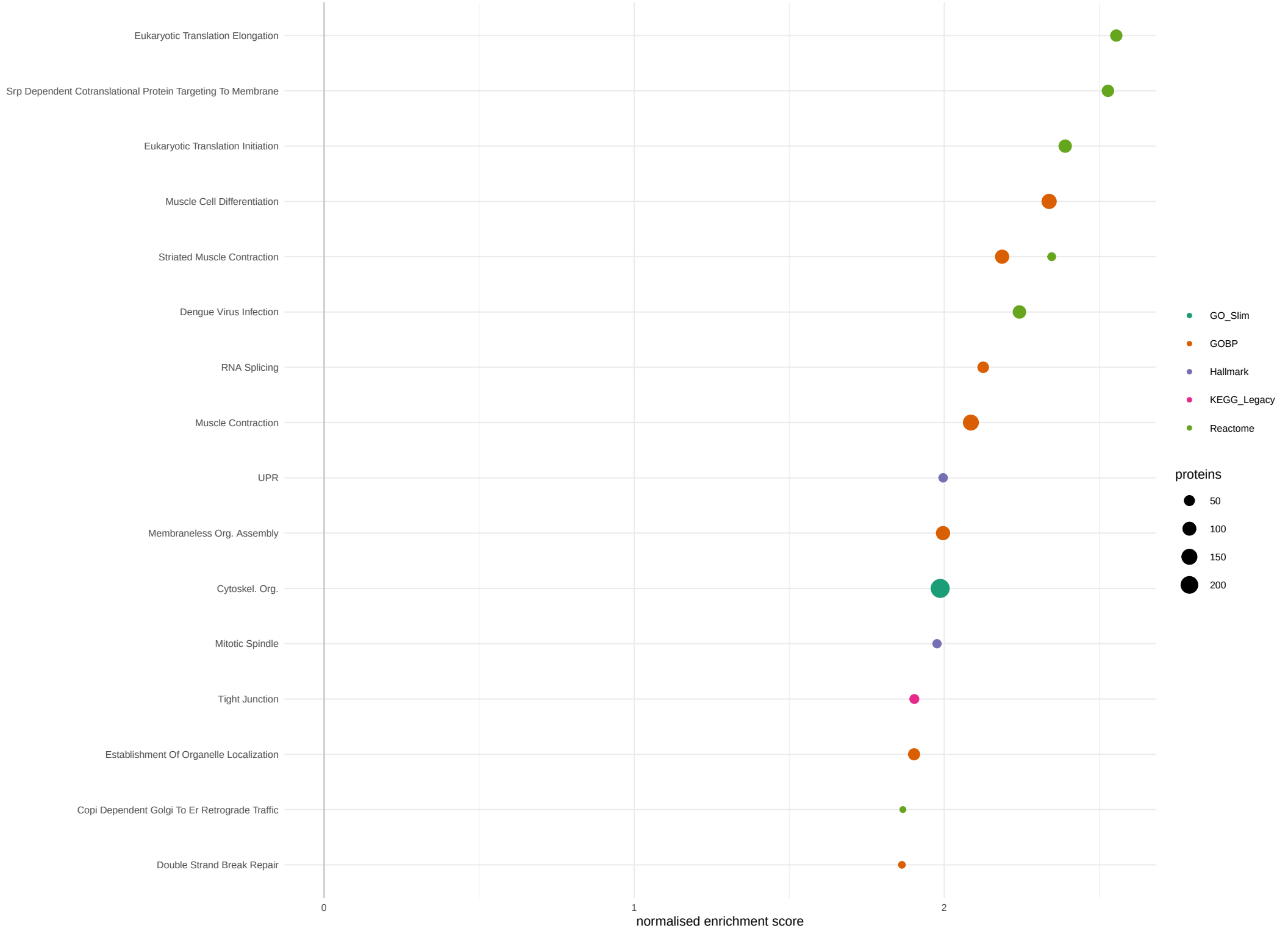

Training_Lo, fgsea collapse survivors at FDR < 0.05
52 sets. Page 1 of 2.
Protein Localization
Mito. Membrane Org.
Osteoblast Differentiation
Sensory Processing Of Sound
Mito. Org.
Sars Cov 2 Infection
Apical Junction
Negative Reg. Secretion
Hemostasis
Actin Filament Org.
Integrin Mediated Signaling
Cell Substrate Adhesion
Cilium Org.
Positive Reg. MAPK Cascade
Protein Localization To Membrane
Mhc Class Ii Antigen Presentation
FA Catabolism
Heme Metabolism
Alcohol Biosynthesis
Glutathione Metabolism
Neuromuscular Process
Cardiac Muscle Contraction
Nuclear Envelope Org.
Mitotic Cell Cycle
Myoblast Differentiation
Cellular Senescence
Reactive Oxygen Species Pathway
Reg. Muscle System Process
M Phase
ARVC
MAPK Signaling
Cell Cycle G1 S Phase Transition
Muscle Organ Development
Calcium Mediated Signaling
Muscle Organ Morphogenesis
Reg. Hemopoiesis
Nuclear Events Mediated By Nfe2l2
Neddylation
Antigen Processing Ubiquitination Proteasome Degradation
C Type Lectin Receptors Clrs
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.


Training_Lo, fgsea collapse survivors at FDR < 0.05
52 sets. Page 2 of 2.
Resp. ETC
OXPHOS
Huntingtons Disease
Mito. Protein Degradation
Parkinsons Disease
Mito. Gene Expression
Mito. Translation Elongation
Eph Ephrin Signaling
Reg. Cdh1 Posttranslational Processing & Trafficking To Plasma Membrane
Protein Localization To Mito.
Gen. of Precursor Metabolites
Complex Iii Assembly
0.0
0.5
1.0
1.5
2.0
normalised enrichment score
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

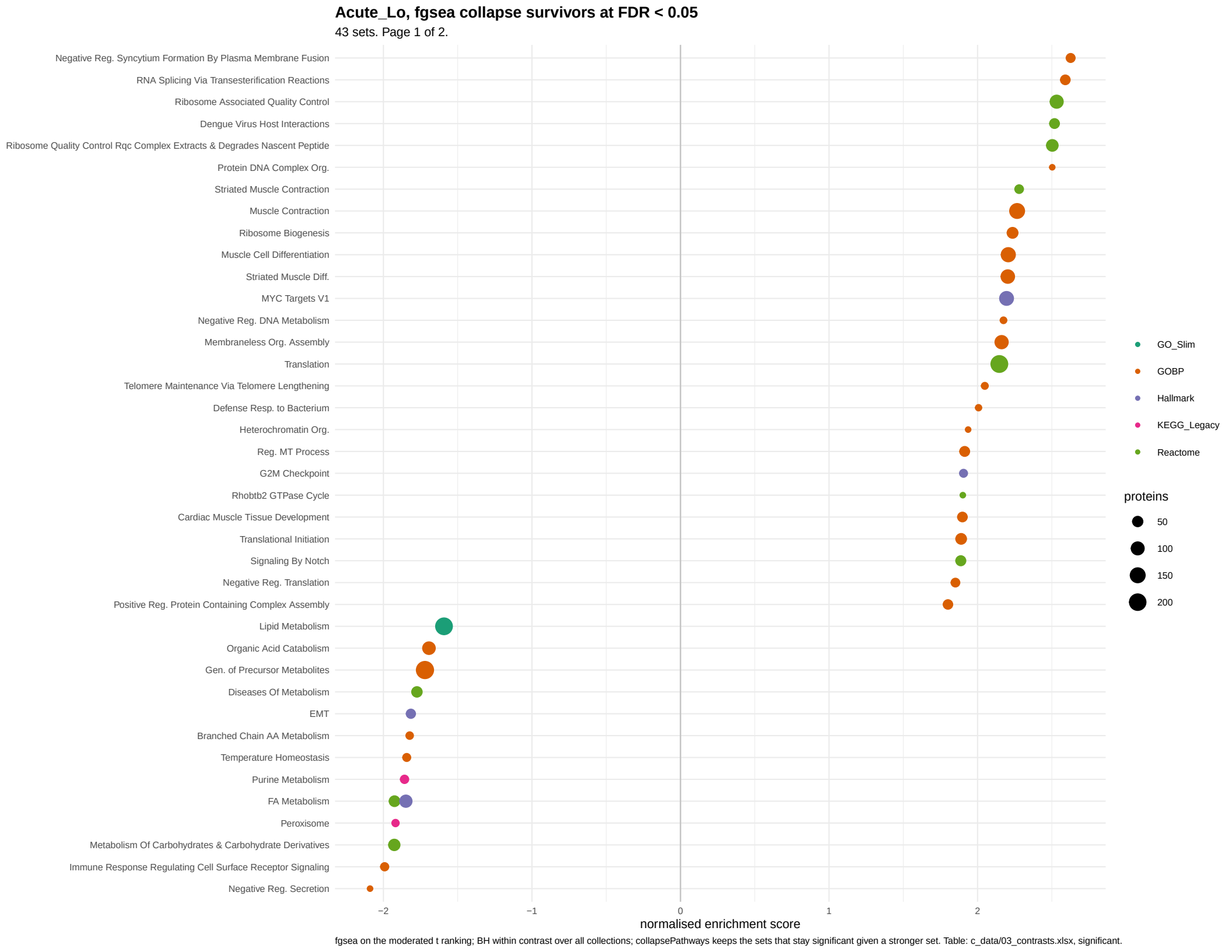

Acute_Hi, fgsea collapse survivors at FDR < 0.05
57 sets. Page 1 of 2.
Sars Cov Infections
Erad Pathway
Vesicle Budding From Membrane
Resp. to Calcium Ion
Cellular Resp. to Biotic Stimulus
Rho GTPase Effectors
N−Linked Glycosylation
Viral Process
Cell Differentiation
Protein RNA Complex Org.
Reg. DNA−Templated Transcription
Vesicle−Mediated Transport
Programmed Cell Death
Antigen Processing Ubiquitination Proteasome Degradation
Mito. Org.
Resp. to Organophosphorus
Glycolysis Gluconeogenesis
Pyruvate Metabolism
Glycolysis
Ketone Biosynthesis
Mito. Gene Expression
Sensory Perception Of Mechanical Stimulus
Cytochrome Complex Assembly
Modified AA Metabolism
Mito. Biogenesis
Nucleobase−Containing Small Molecule Metabolism
Peroxisomal Protein Import
Mito. FA Beta Oxidation
Protein Localization
Gen. of Precursor Metabolites
Sulfur Compound Metabolism
OXPHOS
Positive Reg. Interleukin 2 Production
Heme Metabolism
Dicarboxylic Acid Metabolism
FA Metabolism
Organic Acid Catabolism
TCA Cycle
Aerobic Resp. + ETC
Tricarboxylic Acid Cycle
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
100
200
300
400
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

Acute_Hi, fgsea collapse survivors at FDR < 0.05

57 sets. Page 2 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

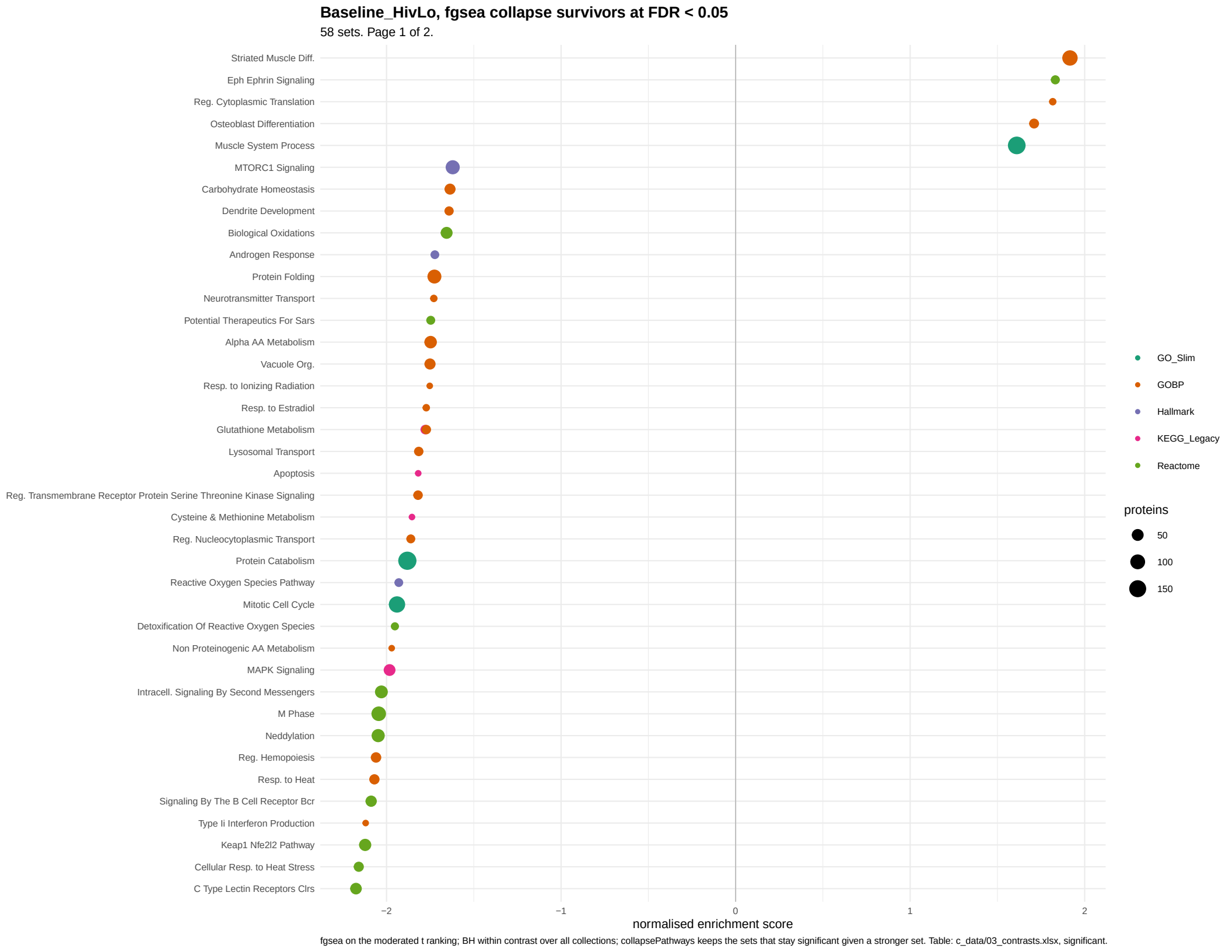

Acute_Lo, fgsea collapse survivors at FDR < 0.05
43 sets. Page 1 of 2.
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
RNA Splicing Via Transesterification Reactions
Ribosome Associated Quality Control
Dengue Virus Host Interactions
Ribosome Quality Control Rqc Complex Extracts & Degrades Nascent Peptide
Protein DNA Complex Org.
Striated Muscle Contraction
Muscle Contraction
Ribosome Biogenesis
Muscle Cell Differentiation
Striated Muscle Diff.
MYC Targets V1
Negative Reg. DNA Metabolism
Membraneless Org. Assembly
Translation
Telomere Maintenance Via Telomere Lengthening
Defense Resp. to Bacterium
Heterochromatin Org.
Reg. MT Process
G2M Checkpoint
Rhobtb2 GTPase Cycle
Cardiac Muscle Tissue Development
Translational Initiation
Signaling By Notch
Negative Reg. Translation
Positive Reg. Protein Containing Complex Assembly
Lipid Metabolism
Organic Acid Catabolism
Gen. of Precursor Metabolites
Diseases Of Metabolism
EMT
Branched Chain AA Metabolism
Temperature Homeostasis
Purine Metabolism
FA Metabolism
Peroxisome
Metabolism Of Carbohydrates & Carbohydrate Derivatives
Immune Response Regulating Cell Surface Receptor Signaling
Negative Reg. Secretion
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

Acute_Lo, fgsea collapse survivors at FDR < 0.05

43 sets. Page 2 of 2.



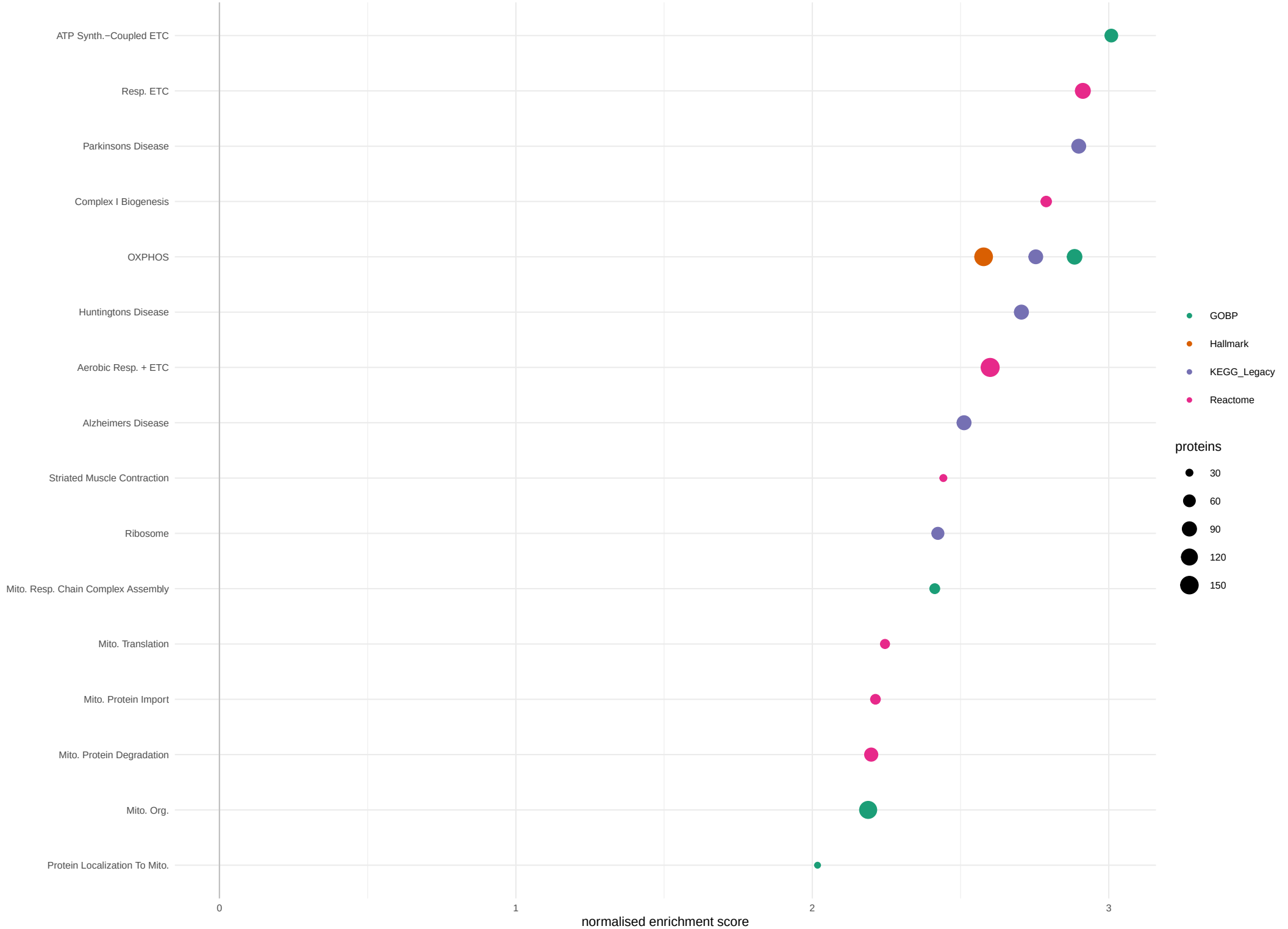
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.



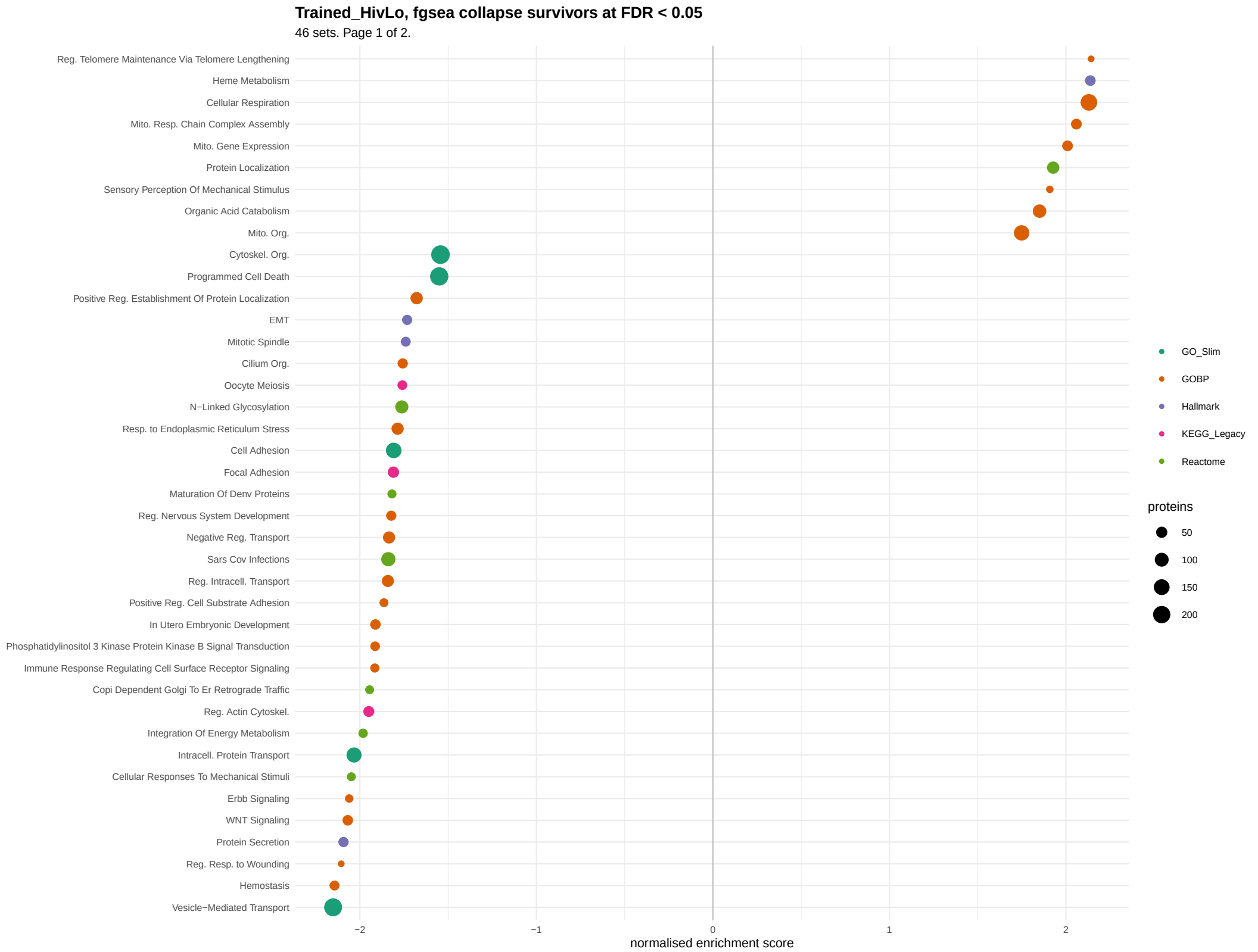
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

Baseline_HivLo, fgsea collapse survivors at FDR < 0.05

58 sets. Page 2 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.



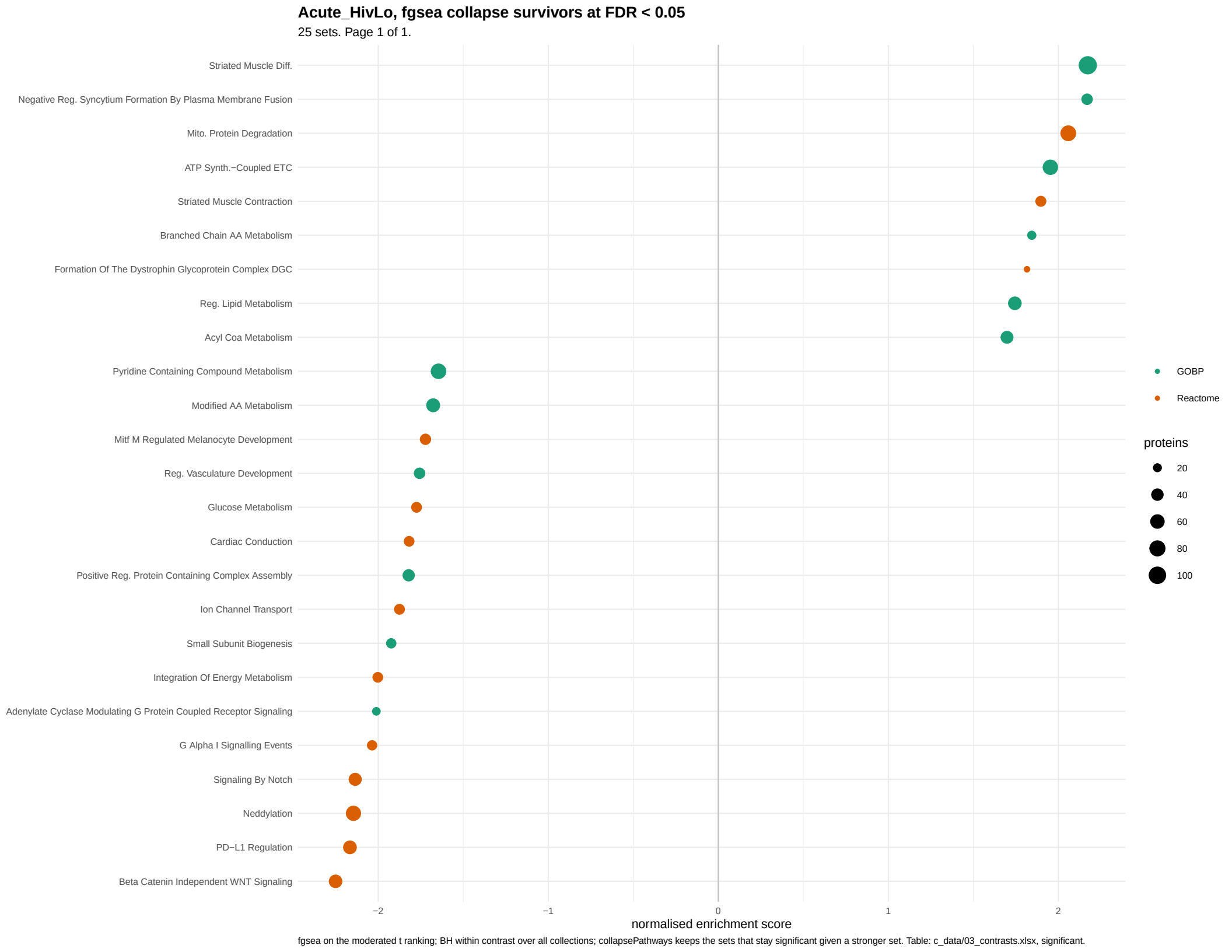
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

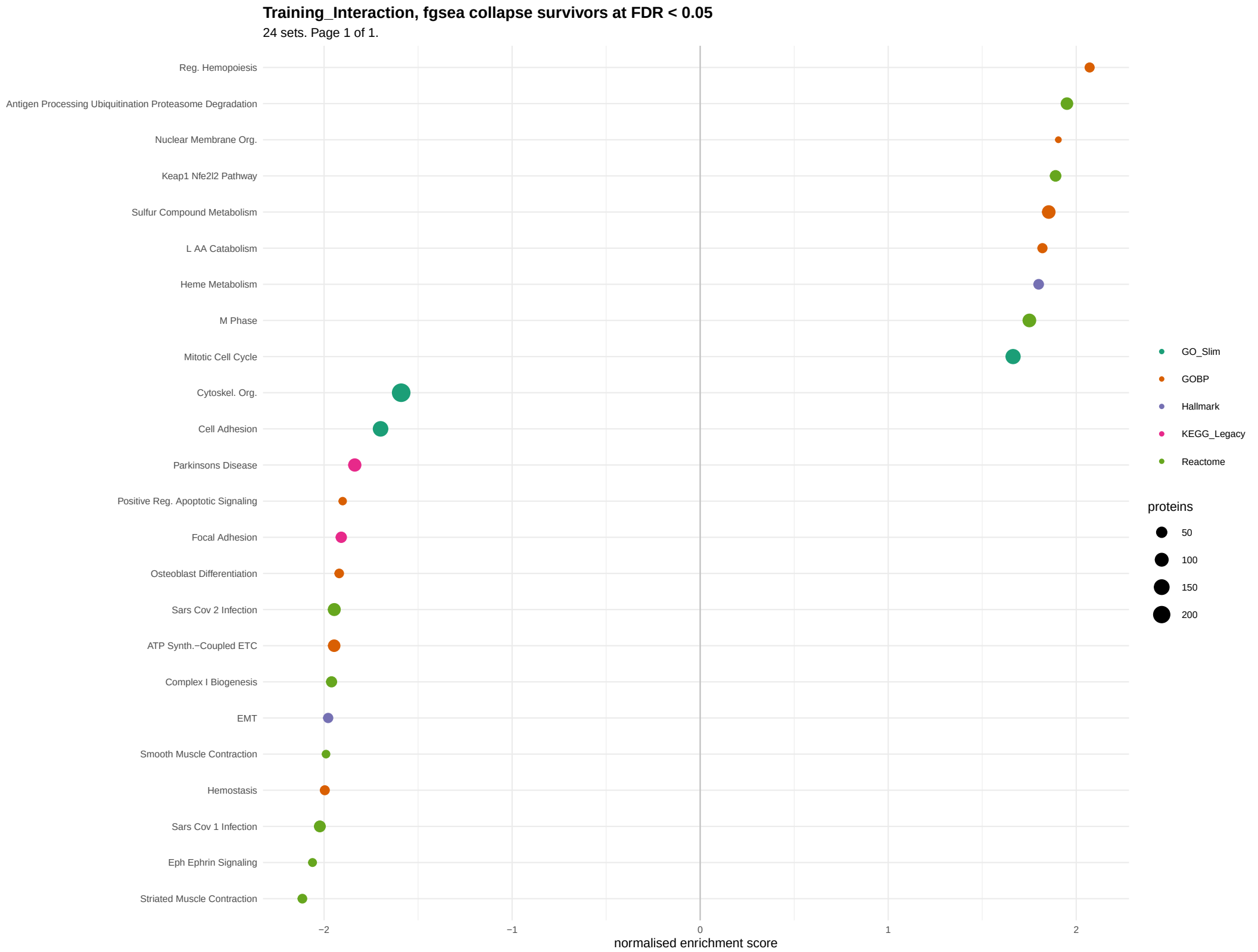
Trained_HivLo, fgsea collapse survivors at FDR < 0.05

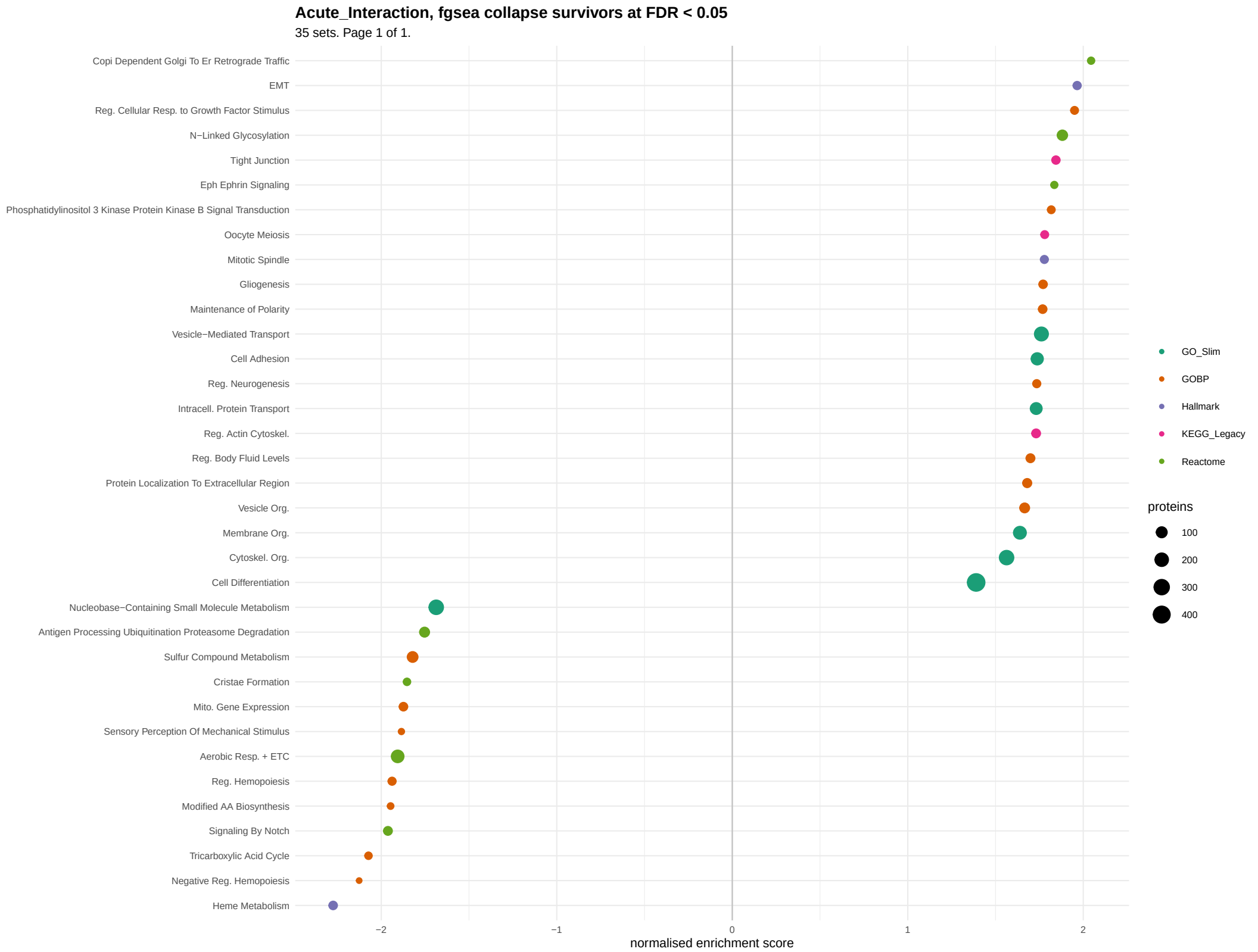
46 sets. Page 2 of 2.



fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

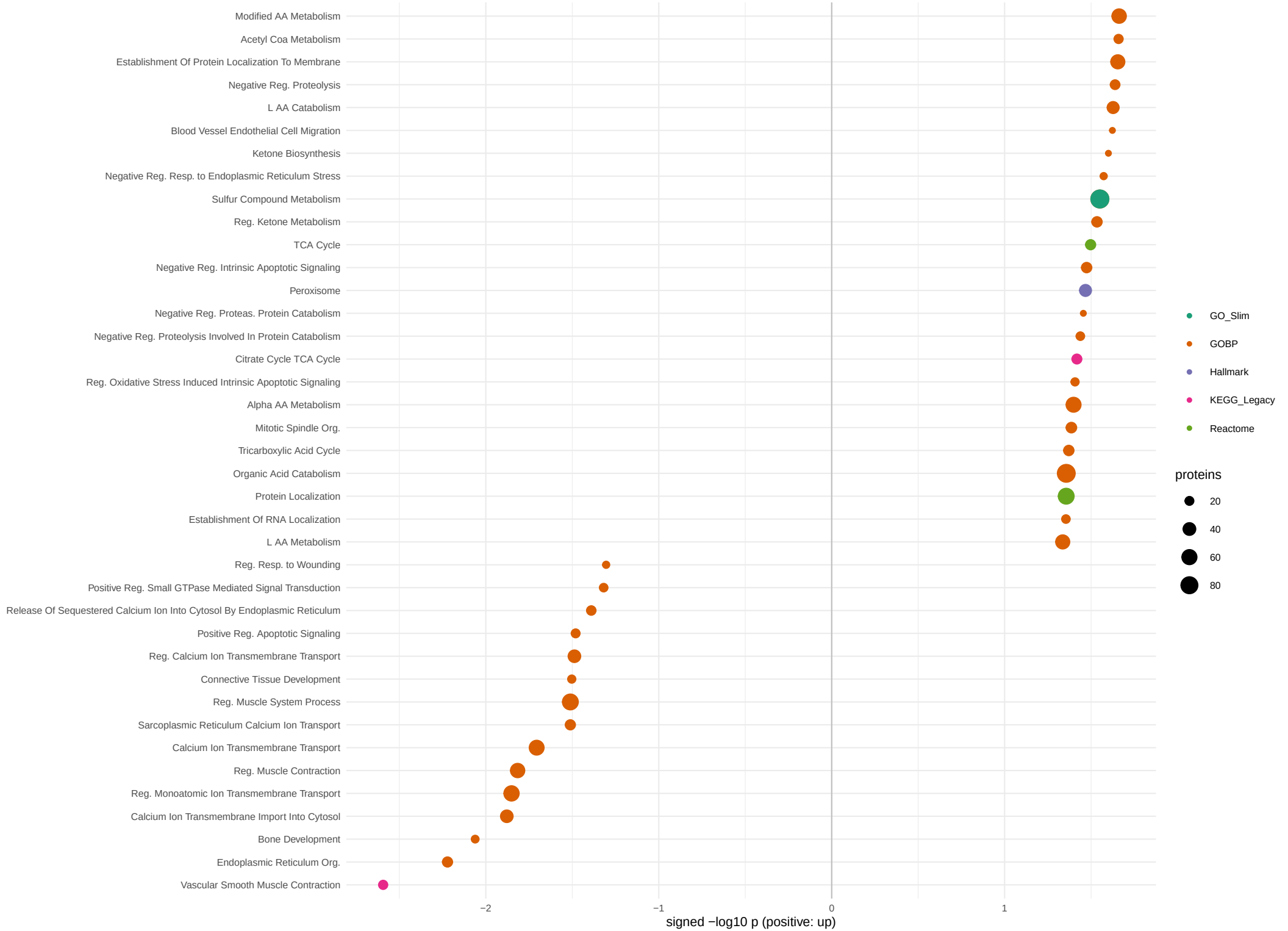

Acute_HivLo, fgsea collapse survivors at FDR < 0.05
25 sets. Page 1 of 1.
Striated Muscle Diff.
Negative Reg. Syncytium Formation By Plasma Membrane Fusion
Mito. Protein Degradation
ATP Synth.−Coupled ETC
Striated Muscle Contraction
Branched Chain AA Metabolism
Formation Of The Dystrophin Glycoprotein Complex DGC
Reg. Lipid Metabolism
Acyl Coa Metabolism
Pyridine Containing Compound Metabolism
Modified AA Metabolism
Mitf M Regulated Melanocyte Development
Reg. Vasculature Development
Glucose Metabolism
Cardiac Conduction
Positive Reg. Protein Containing Complex Assembly
Ion Channel Transport
Small Subunit Biogenesis
Integration Of Energy Metabolism
Adenylate Cyclase Modulating G Protein Coupled Receptor Signaling
G Alpha I Signalling Events
Signaling By Notch
Neddylation
PD−L1 Regulation
Beta Catenin Independent WNT Signaling
−2
−1
0
1
2
normalised enrichment score
GOBP
Reactome
proteins
20
40
60
80
100
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.


Training_Interaction, fgsea collapse survivors at FDR < 0.05
24 sets. Page 1 of 1.
Reg. Hemopoiesis
Antigen Processing Ubiquitination Proteasome Degradation
Nuclear Membrane Org.
Keap1 Nfe2l2 Pathway
Sulfur Compound Metabolism
L AA Catabolism
Heme Metabolism
M Phase
Mitotic Cell Cycle
Cytoskel. Org.
Cell Adhesion
Parkinsons Disease
Positive Reg. Apoptotic Signaling
Focal Adhesion
Osteoblast Differentiation
Sars Cov 2 Infection
ATP Synth.−Coupled ETC
Complex I Biogenesis
EMT
Smooth Muscle Contraction
Hemostasis
Sars Cov 1 Infection
Eph Ephrin Signaling
Striated Muscle Contraction
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

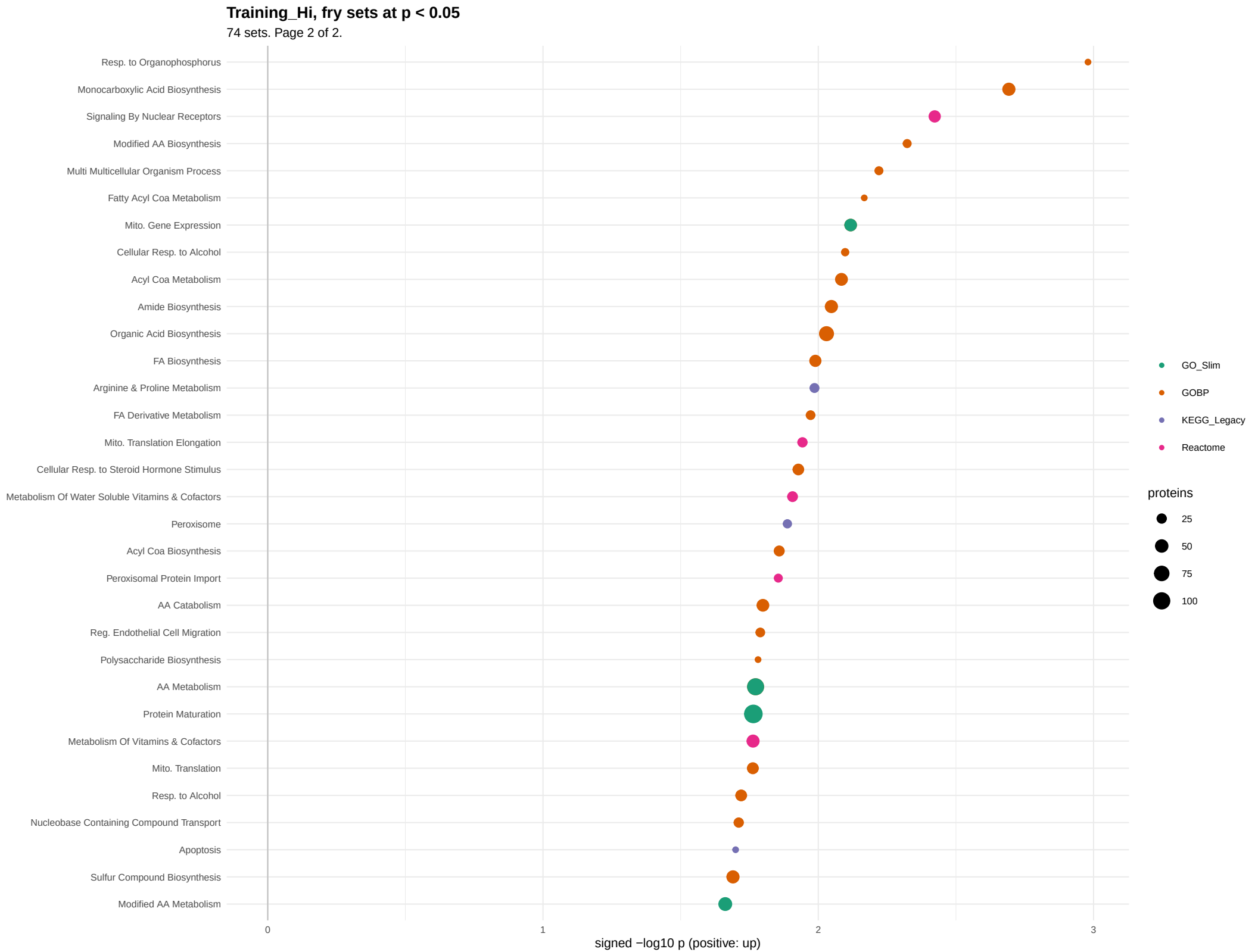

Acute_Interaction, fgsea collapse survivors at FDR < 0.05
35 sets. Page 1 of 1.
Copi Dependent Golgi To Er Retrograde Traffic
EMT
Reg. Cellular Resp. to Growth Factor Stimulus
N−Linked Glycosylation
Tight Junction
Eph Ephrin Signaling
Phosphatidylinositol 3 Kinase Protein Kinase B Signal Transduction
Oocyte Meiosis
Mitotic Spindle
Gliogenesis
Maintenance of Polarity
Vesicle−Mediated Transport
Cell Adhesion
Reg. Neurogenesis
Intracell. Protein Transport
Reg. Actin Cytoskel.
Reg. Body Fluid Levels
Protein Localization To Extracellular Region
Vesicle Org.
Membrane Org.
Cytoskel. Org.
Cell Differentiation
Nucleobase−Containing Small Molecule Metabolism
Antigen Processing Ubiquitination Proteasome Degradation
Sulfur Compound Metabolism
Cristae Formation
Mito. Gene Expression
Sensory Perception Of Mechanical Stimulus
Aerobic Resp. + ETC
Reg. Hemopoiesis
Modified AA Biosynthesis
Signaling By Notch
Tricarboxylic Acid Cycle
Negative Reg. Hemopoiesis
Heme Metabolism
−2
−1
0
1
2
normalised enrichment score
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
100
200
300
400
fgsea on the moderated t ranking; BH within contrast over all collections; collapsePathways keeps the sets that stay significant given a stronger set. Table: c_data/03_contrasts.xlsx, significant.

Training_Hi, fry sets at p < 0.05

74 sets. Page 1 of 2.



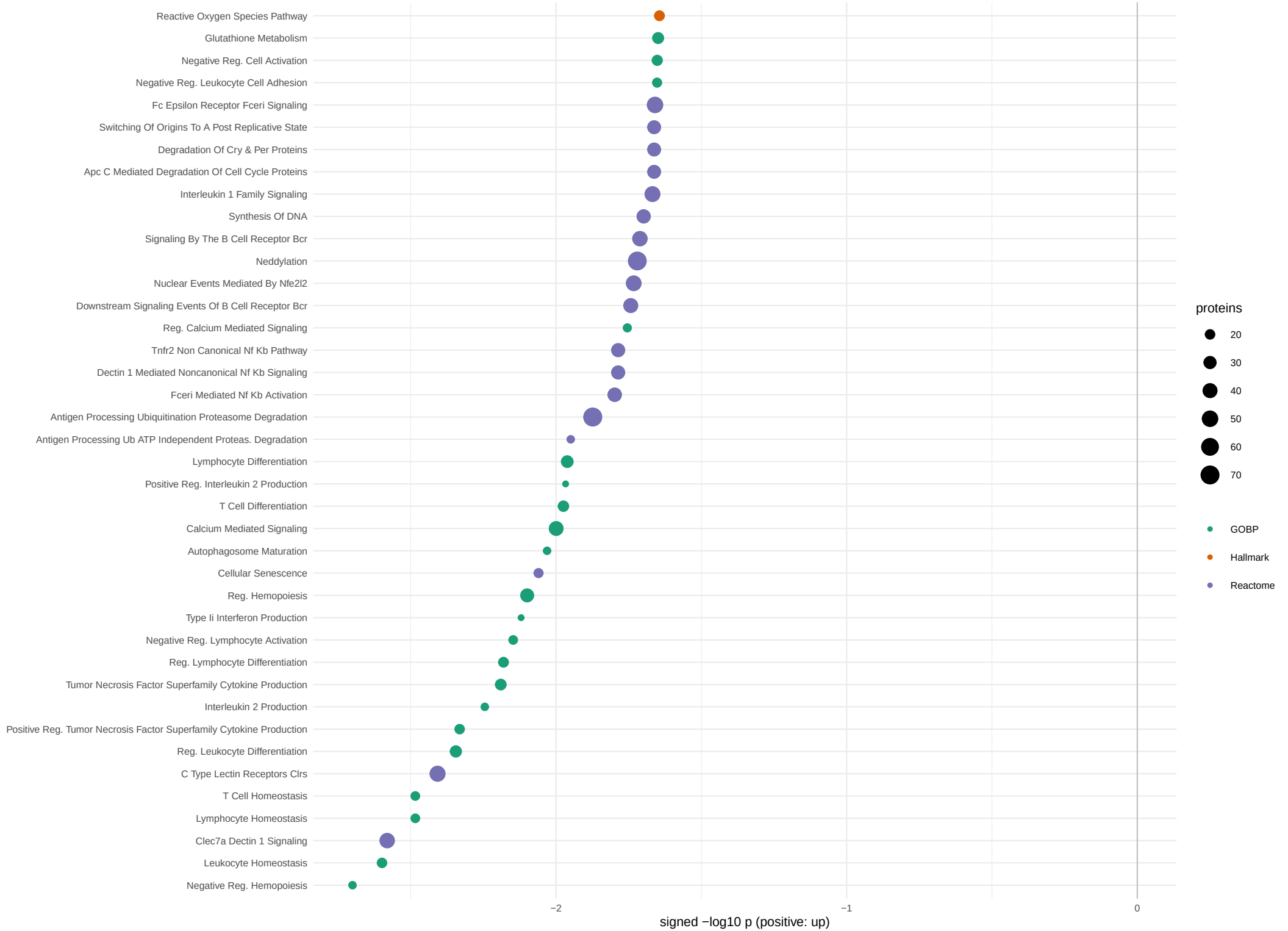
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



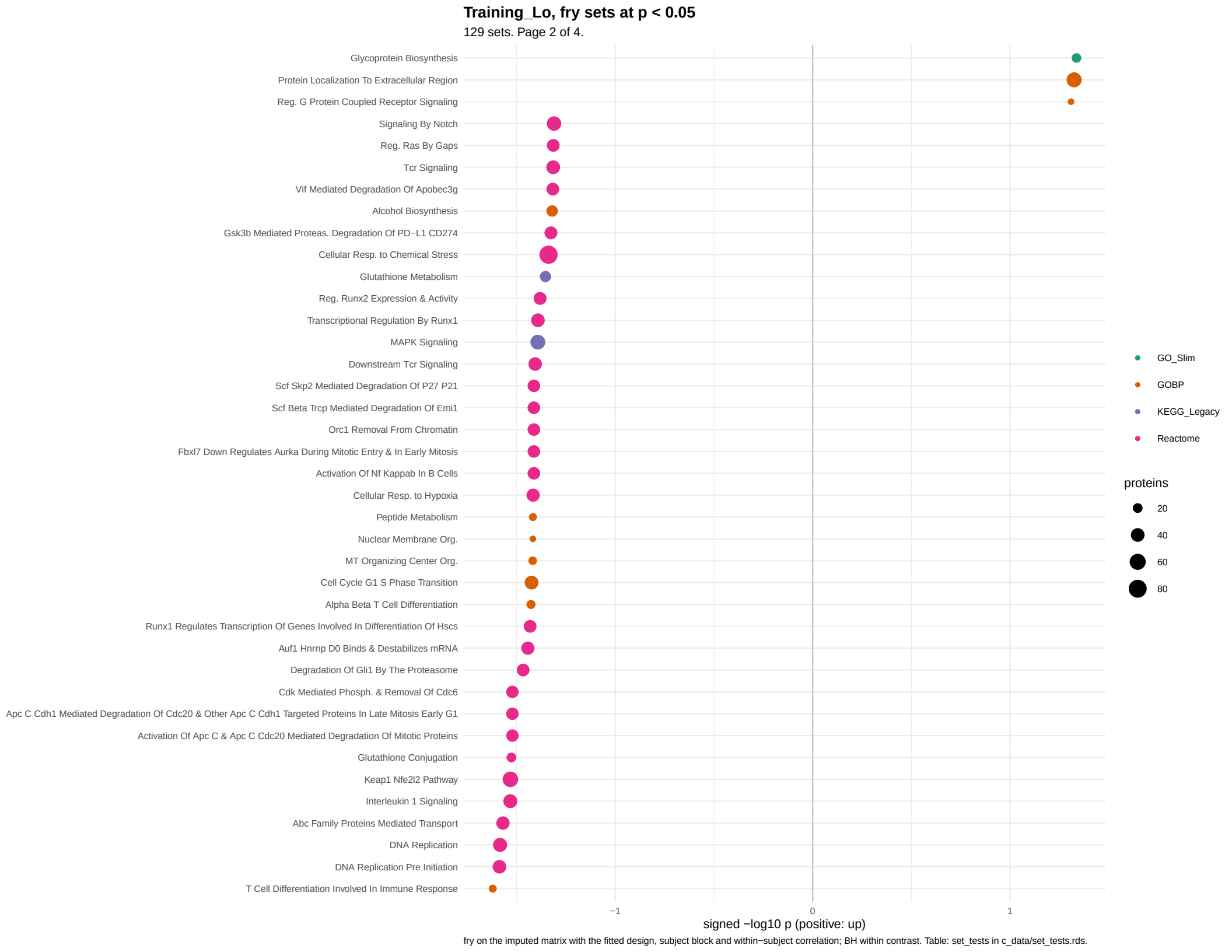
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

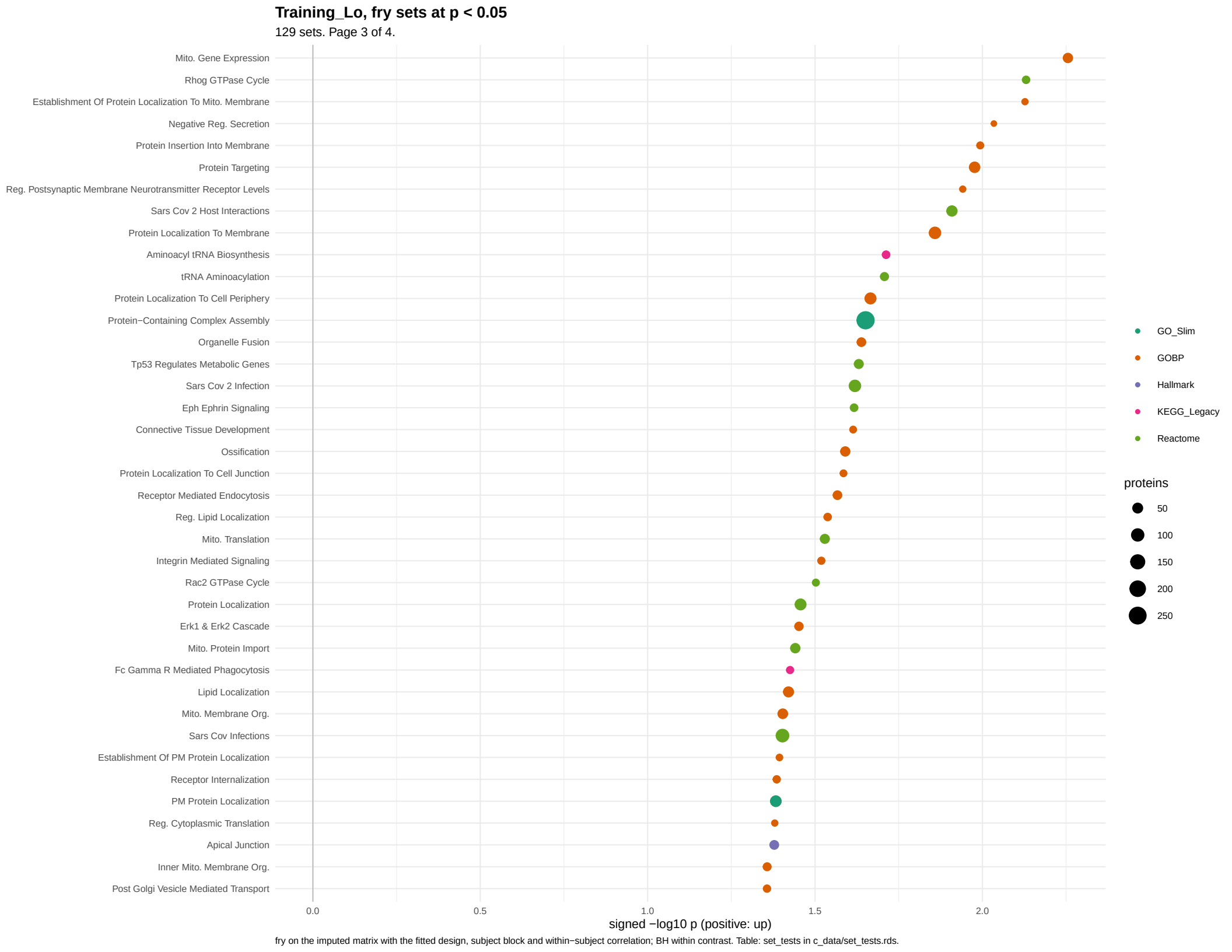
Training_Lo, fry sets at p < 0.05

129 sets. Page 1 of 4.



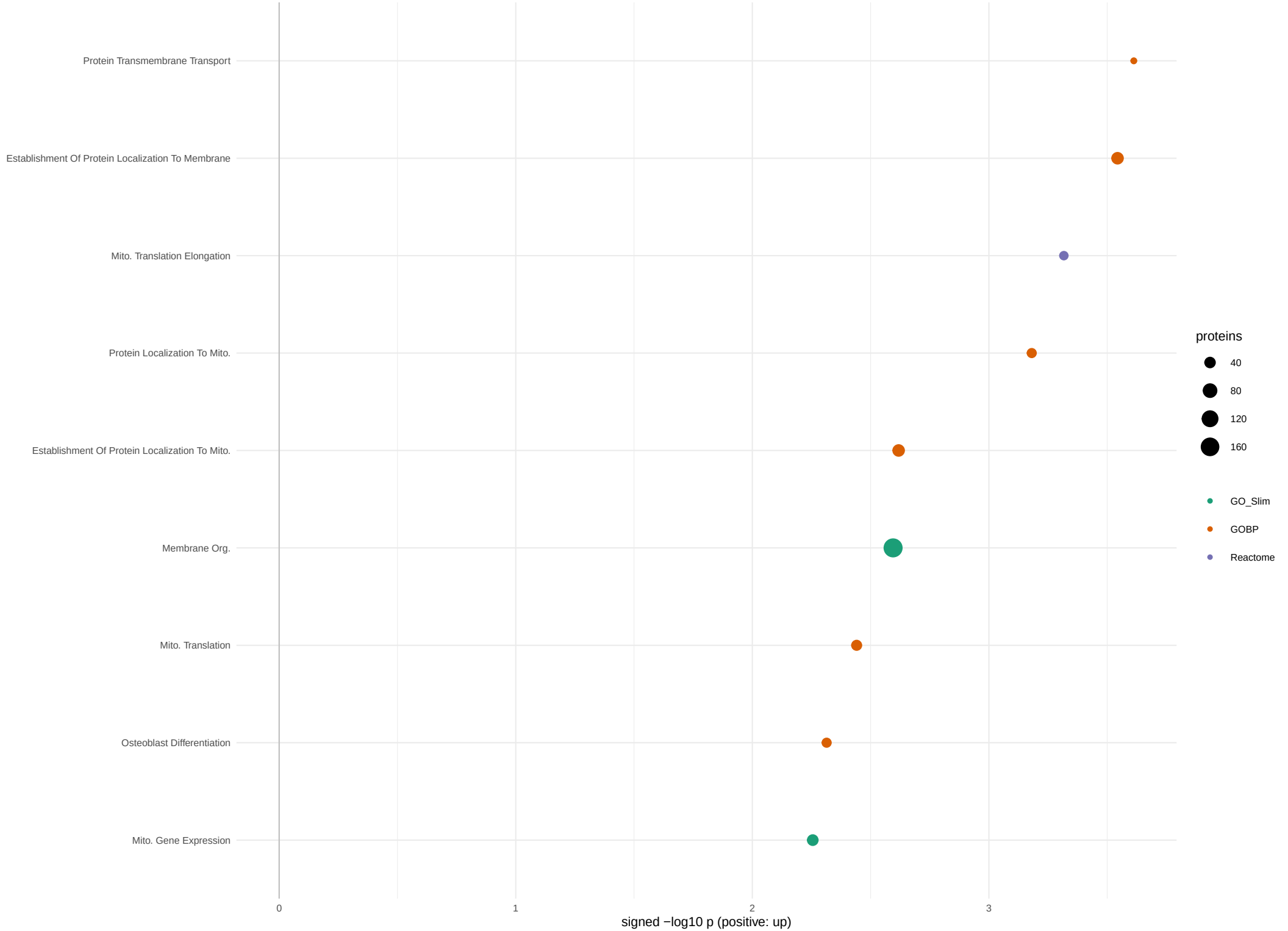
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Training_Lo, fry sets at p < 0.05
129 sets. Page 2 of 4.
Glycoprotein Biosynthesis
Protein Localization To Extracellular Region
Reg. G Protein Coupled Receptor Signaling
Signaling By Notch
Reg. Ras By Gaps
Tcr Signaling
Vif Mediated Degradation Of Apobec3g
Alcohol Biosynthesis
Gsk3b Mediated Proteas. Degradation Of PD−L1 CD274
Cellular Resp. to Chemical Stress
Glutathione Metabolism
Reg. Runx2 Expression & Activity
Transcriptional Regulation By Runx1
MAPK Signaling
Downstream Tcr Signaling
Scf Skp2 Mediated Degradation Of P27 P21
Scf Beta Trcp Mediated Degradation Of Emi1
Orc1 Removal From Chromatin
Fbxl7 Down Regulates Aurka During Mitotic Entry & In Early Mitosis
Activation Of Nf Kappab In B Cells
Cellular Resp. to Hypoxia
Peptide Metabolism
Nuclear Membrane Org.
MT Organizing Center Org.
Cell Cycle G1 S Phase Transition
Alpha Beta T Cell Differentiation
Runx1 Regulates Transcription Of Genes Involved In Differentiation Of Hscs
Auf1 Hnrnp D0 Binds & Destabilizes mRNA
Degradation Of Gli1 By The Proteasome
Cdk Mediated Phosph. & Removal Of Cdc6
Apc C Cdh1 Mediated Degradation Of Cdc20 & Other Apc C Cdh1 Targeted Proteins In Late Mitosis Early G1
Activation Of Apc C & Apc C Cdc20 Mediated Degradation Of Mitotic Proteins
Glutathione Conjugation
Keap1 Nfe2l2 Pathway
Interleukin 1 Signaling
Abc Family Proteins Mediated Transport
DNA Replication
DNA Replication Pre Initiation
T Cell Differentiation Involved In Immune Response
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
KEGG_Legacy
Reactome
proteins
20
40
60
80
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

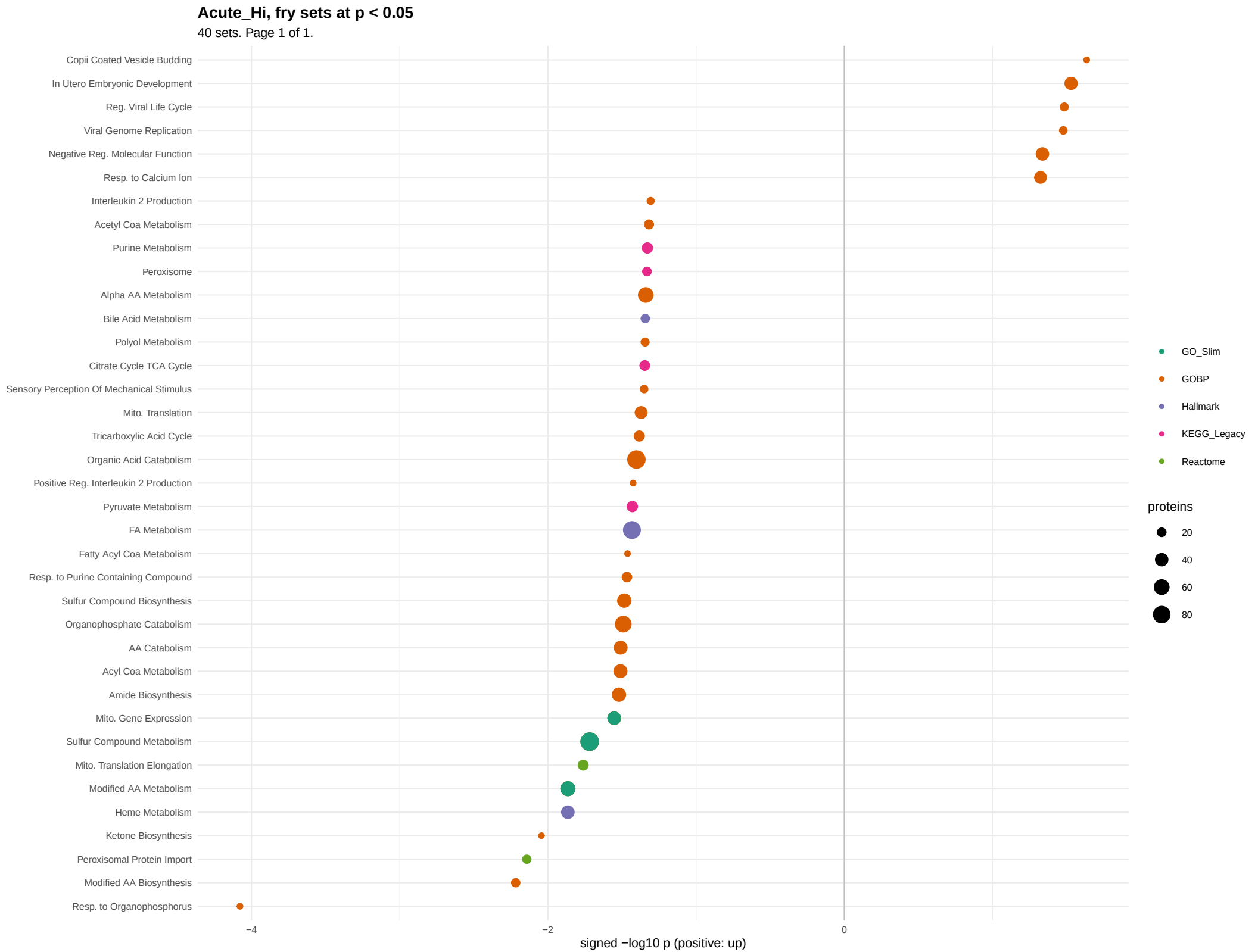

Training_Lo, fry sets at p < 0.05
129 sets. Page 3 of 4.
Mito. Gene Expression
Rhog GTPase Cycle
Establishment Of Protein Localization To Mito. Membrane
Negative Reg. Secretion
Protein Insertion Into Membrane
Protein Targeting
Reg. Postsynaptic Membrane Neurotransmitter Receptor Levels
Sars Cov 2 Host Interactions
Protein Localization To Membrane
Aminoacyl tRNA Biosynthesis
tRNA Aminoacylation
Protein Localization To Cell Periphery
Protein−Containing Complex Assembly
Organelle Fusion
Tp53 Regulates Metabolic Genes
Sars Cov 2 Infection
Eph Ephrin Signaling
Connective Tissue Development
Ossification
Protein Localization To Cell Junction
Receptor Mediated Endocytosis
Reg. Lipid Localization
Mito. Translation
Integrin Mediated Signaling
Rac2 GTPase Cycle
Protein Localization
Erk1 & Erk2 Cascade
Mito. Protein Import
Fc Gamma R Mediated Phagocytosis
Lipid Localization
Mito. Membrane Org.
Sars Cov Infections
Establishment Of PM Protein Localization
Receptor Internalization
PM Protein Localization
Reg. Cytoplasmic Translation
Apical Junction
Inner Mito. Membrane Org.
Post Golgi Vesicle Mediated Transport
0.0
0.5
1.0
1.5
2.0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
250
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

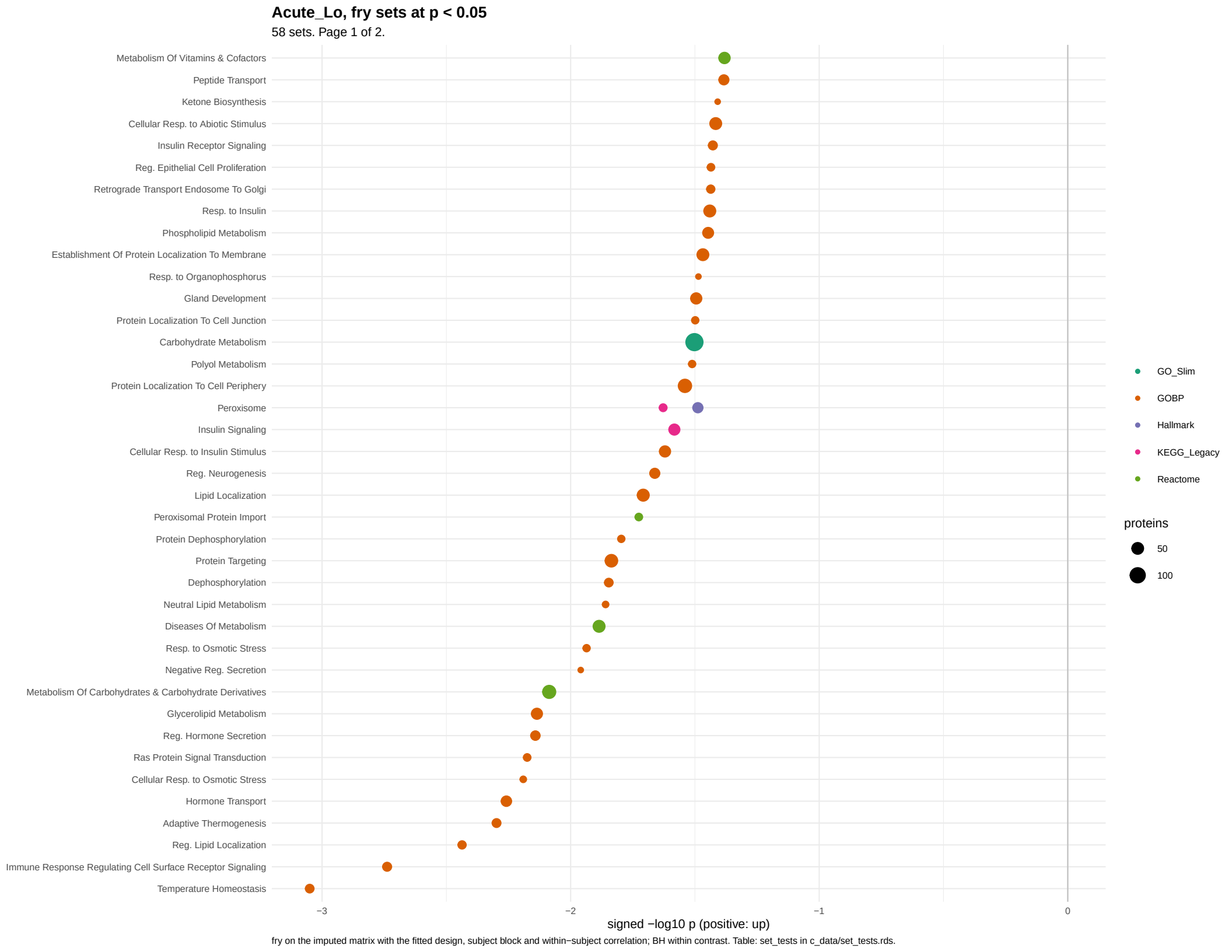
Training_Lo, fry sets at p < 0.05

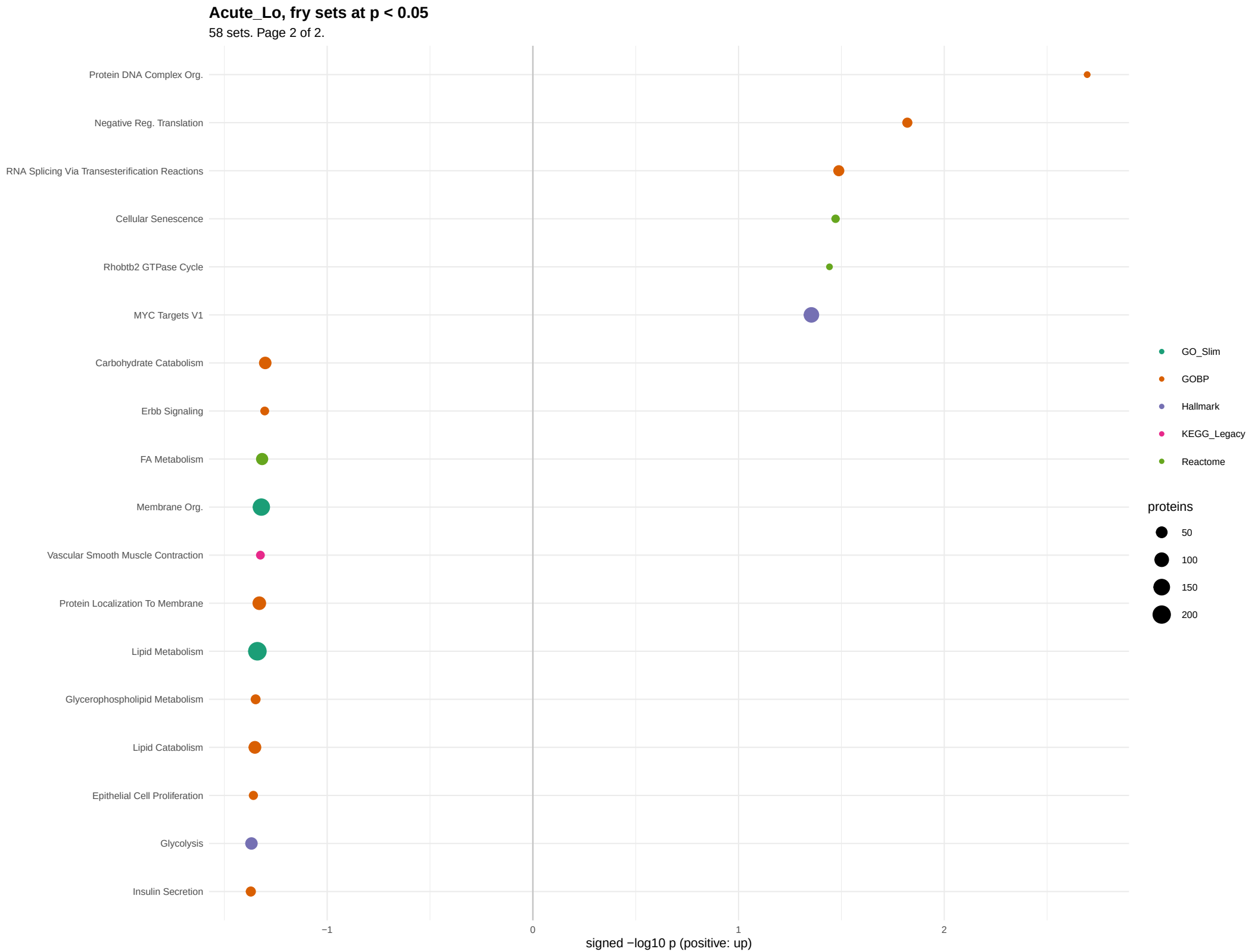
129 sets. Page 4 of 4.

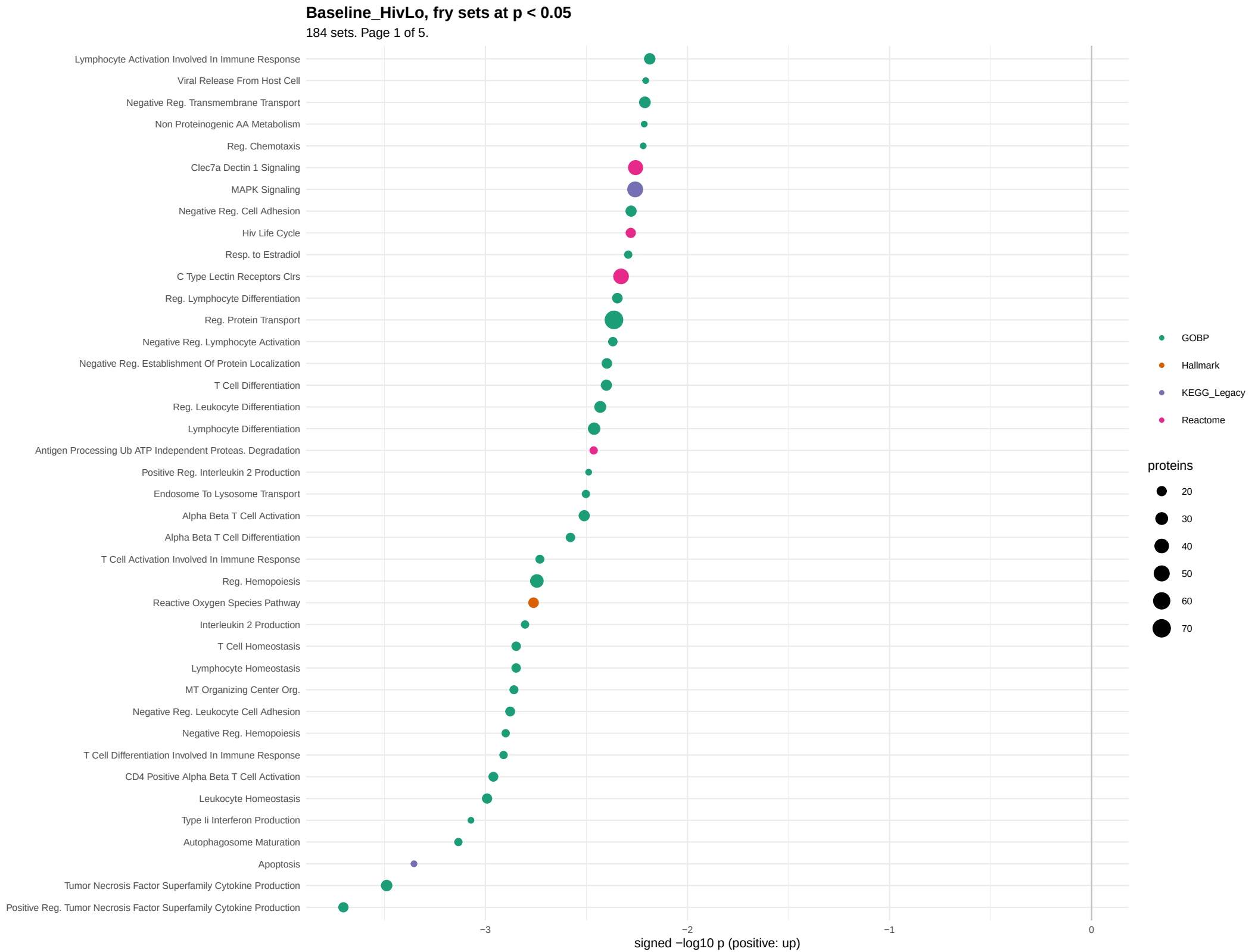


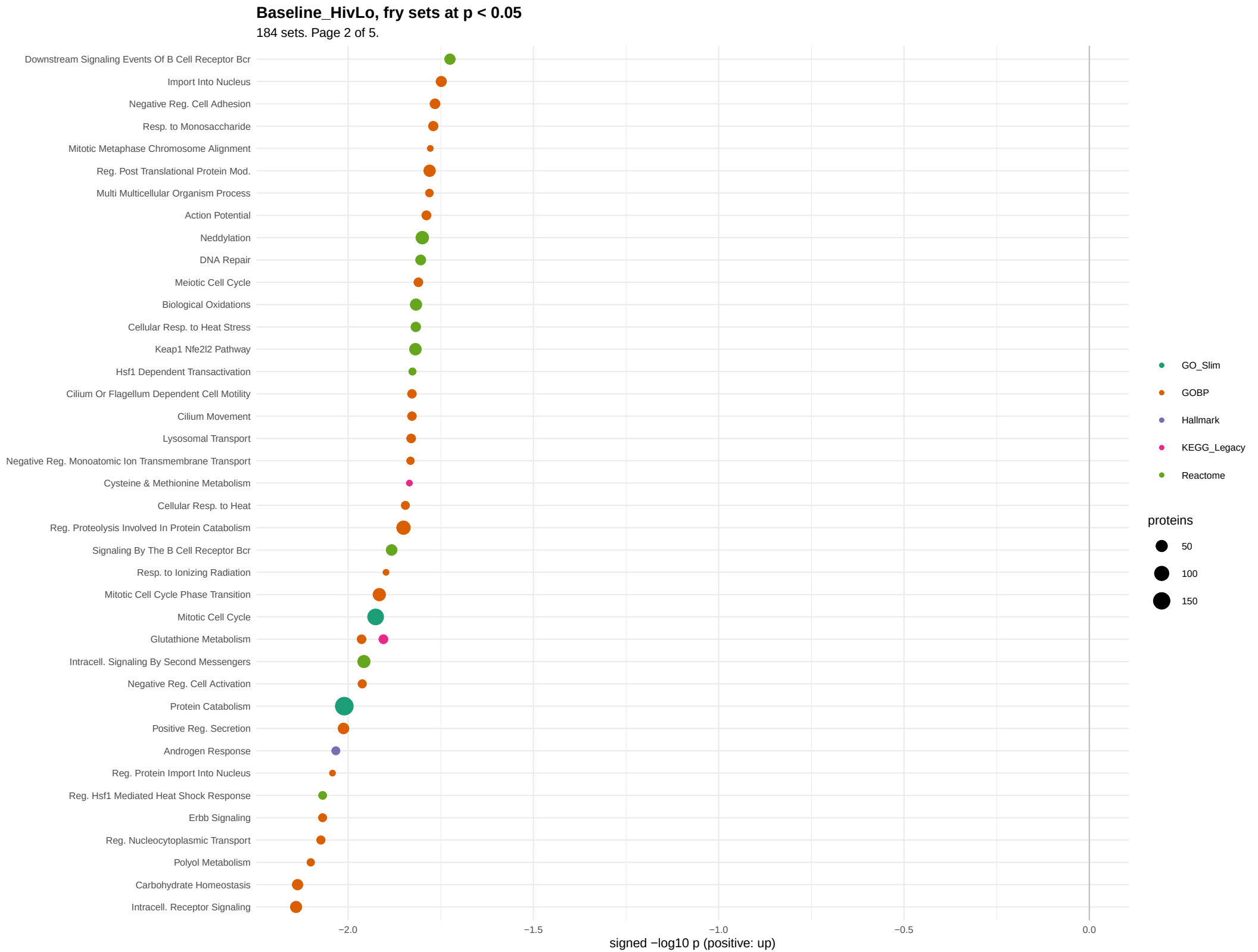
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Hi, fry sets at p < 0.05
40 sets. Page 1 of 1.
Copii Coated Vesicle Budding
In Utero Embryonic Development
Reg. Viral Life Cycle
Viral Genome Replication
Negative Reg. Molecular Function
Resp. to Calcium Ion
Interleukin 2 Production
Acetyl Coa Metabolism
Purine Metabolism
Peroxisome
Alpha AA Metabolism
Bile Acid Metabolism
Polyol Metabolism
Citrate Cycle TCA Cycle
Sensory Perception Of Mechanical Stimulus
Mito. Translation
Tricarboxylic Acid Cycle
Organic Acid Catabolism
Positive Reg. Interleukin 2 Production
Pyruvate Metabolism
FA Metabolism
Fatty Acyl Coa Metabolism
Resp. to Purine Containing Compound
Sulfur Compound Biosynthesis
Organophosphate Catabolism
AA Catabolism
Acyl Coa Metabolism
Amide Biosynthesis
Mito. Gene Expression
Sulfur Compound Metabolism
Mito. Translation Elongation
Modified AA Metabolism
Heme Metabolism
Ketone Biosynthesis
Peroxisomal Protein Import
Modified AA Biosynthesis
Resp. to Organophosphorus
−4
−2
0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
20
40
60
80
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

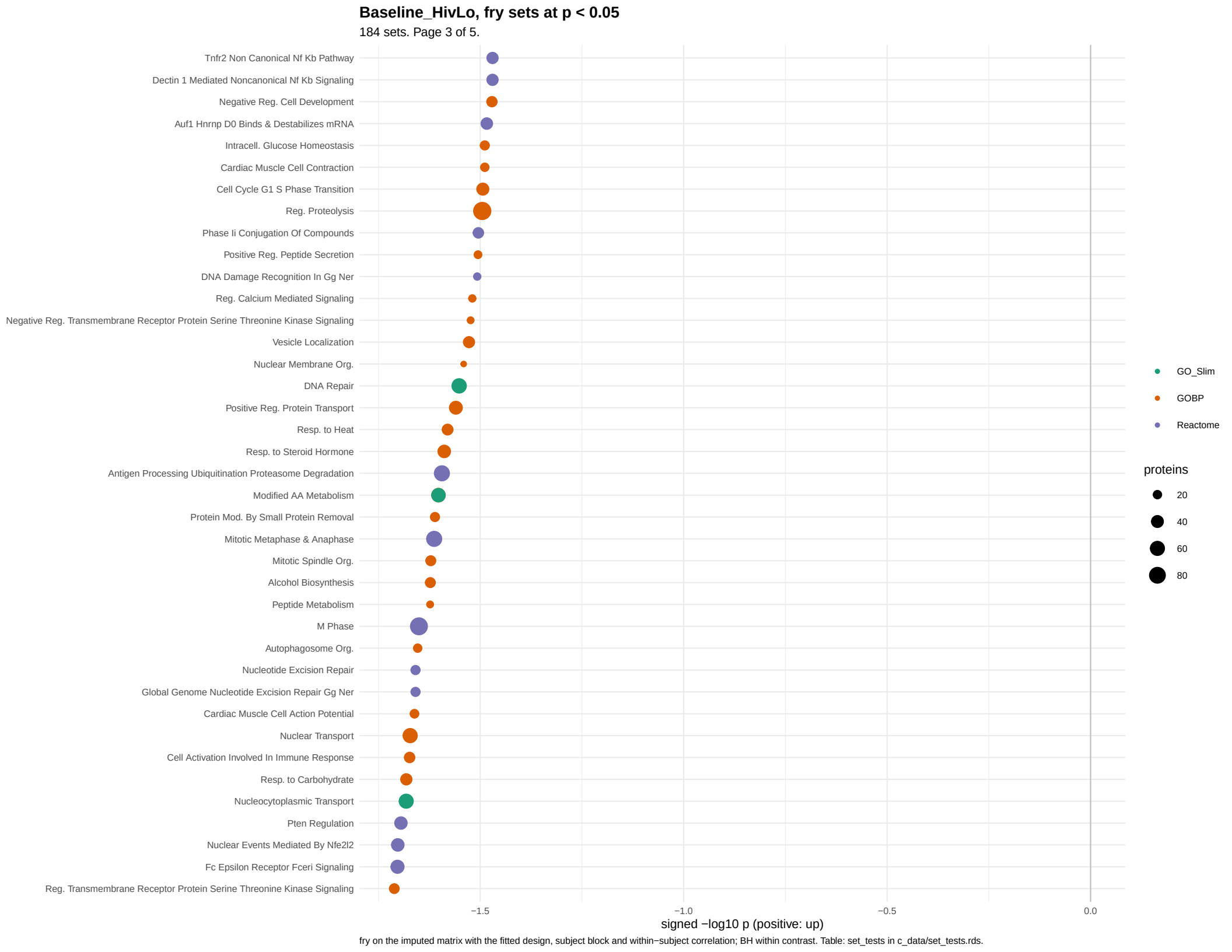

Acute_Lo, fry sets at p < 0.05
58 sets. Page 1 of 2.
Metabolism Of Vitamins & Cofactors
Peptide Transport
Ketone Biosynthesis
Cellular Resp. to Abiotic Stimulus
Insulin Receptor Signaling
Reg. Epithelial Cell Proliferation
Retrograde Transport Endosome To Golgi
Resp. to Insulin
Phospholipid Metabolism
Establishment Of Protein Localization To Membrane
Resp. to Organophosphorus
Gland Development
Protein Localization To Cell Junction
Carbohydrate Metabolism
Polyol Metabolism
Protein Localization To Cell Periphery
Peroxisome
Insulin Signaling
Cellular Resp. to Insulin Stimulus
Reg. Neurogenesis
Lipid Localization
Peroxisomal Protein Import
Protein Dephosphorylation
Protein Targeting
Dephosphorylation
Neutral Lipid Metabolism
Diseases Of Metabolism
Resp. to Osmotic Stress
Negative Reg. Secretion
Metabolism Of Carbohydrates & Carbohydrate Derivatives
Glycerolipid Metabolism
Reg. Hormone Secretion
Ras Protein Signal Transduction
Cellular Resp. to Osmotic Stress
Hormone Transport
Adaptive Thermogenesis
Reg. Lipid Localization
Immune Response Regulating Cell Surface Receptor Signaling
Temperature Homeostasis
−3
−2
−1
0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

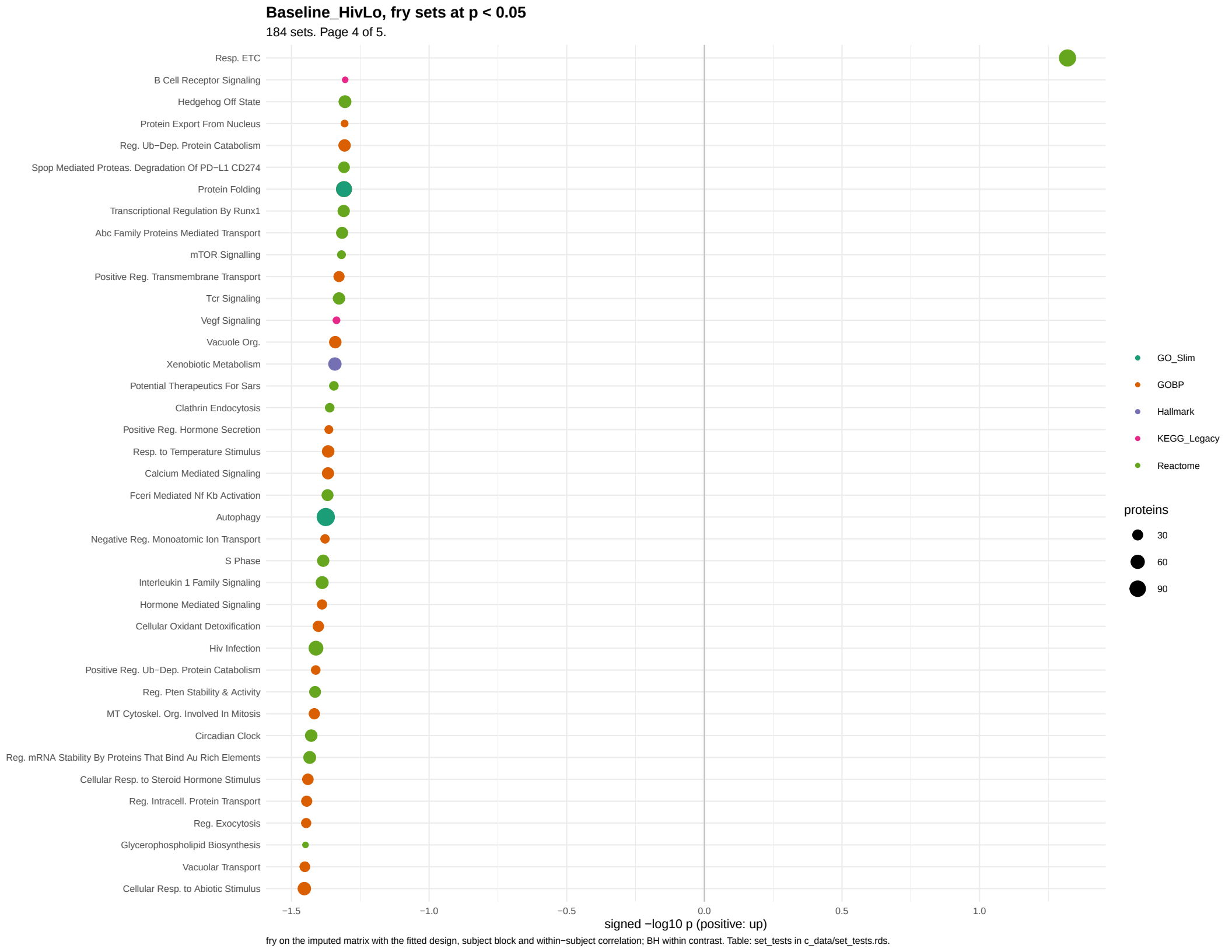

Acute_Lo, fry sets at p < 0.05
58 sets. Page 2 of 2.
Protein DNA Complex Org.
Negative Reg. Translation
RNA Splicing Via Transesterification Reactions
Cellular Senescence
Rhobtb2 GTPase Cycle
MYC Targets V1
Carbohydrate Catabolism
Erbb Signaling
FA Metabolism
Membrane Org.
Vascular Smooth Muscle Contraction
Protein Localization To Membrane
Lipid Metabolism
Glycerophospholipid Metabolism
Lipid Catabolism
Epithelial Cell Proliferation
Glycolysis
Insulin Secretion
−1
0
1
2
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
50
100
150
200
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Baseline_HivLo, fry sets at p < 0.05
184 sets. Page 1 of 5.
Lymphocyte Activation Involved In Immune Response
Viral Release From Host Cell
Negative Reg. Transmembrane Transport
Non Proteinogenic AA Metabolism
Reg. Chemotaxis
Clec7a Dectin 1 Signaling
MAPK Signaling
Negative Reg. Cell Adhesion
Hiv Life Cycle
Resp. to Estradiol
C Type Lectin Receptors Clrs
Reg. Lymphocyte Differentiation
Reg. Protein Transport
Negative Reg. Lymphocyte Activation
Negative Reg. Establishment Of Protein Localization
T Cell Differentiation
Reg. Leukocyte Differentiation
Lymphocyte Differentiation
Antigen Processing Ub ATP Independent Proteas. Degradation
Positive Reg. Interleukin 2 Production
Endosome To Lysosome Transport
Alpha Beta T Cell Activation
Alpha Beta T Cell Differentiation
T Cell Activation Involved In Immune Response
Reg. Hemopoiesis
Reactive Oxygen Species Pathway
Interleukin 2 Production
T Cell Homeostasis
Lymphocyte Homeostasis
MT Organizing Center Org.
Negative Reg. Leukocyte Cell Adhesion
Negative Reg. Hemopoiesis
T Cell Differentiation Involved In Immune Response
CD4 Positive Alpha Beta T Cell Activation
Leukocyte Homeostasis
Type Ii Interferon Production
Autophagosome Maturation
Apoptosis
Tumor Necrosis Factor Superfamily Cytokine Production
Positive Reg. Tumor Necrosis Factor Superfamily Cytokine Production
−3
−2
−1
0
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
20
30
40
50
60
70
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.



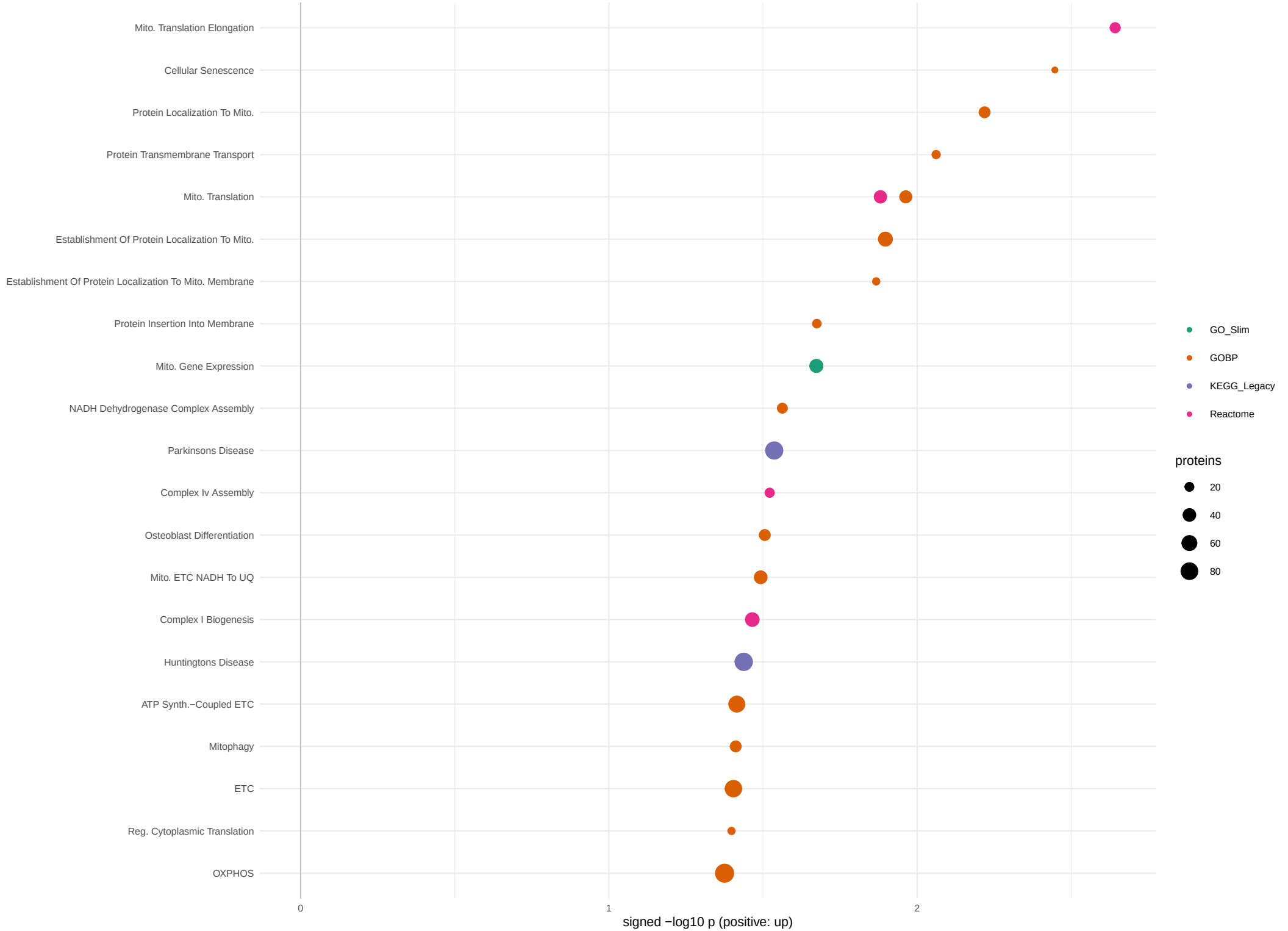
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Baseline_HivLo, fry sets at p < 0.05
184 sets. Page 3 of 5.
Tnfr2 Non Canonical Nf Kb Pathway
Dectin 1 Mediated Noncanonical Nf Kb Signaling
Negative Reg. Cell Development
Auf1 Hnrnp D0 Binds & Destabilizes mRNA
Intracell. Glucose Homeostasis
Cardiac Muscle Cell Contraction
Cell Cycle G1 S Phase Transition
Reg. Proteolysis
Phase Ii Conjugation Of Compounds
Positive Reg. Peptide Secretion
DNA Damage Recognition In Gg Ner
Reg. Calcium Mediated Signaling
Negative Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
Vesicle Localization
Nuclear Membrane Org.
DNA Repair
Positive Reg. Protein Transport
Resp. to Heat
Resp. to Steroid Hormone
Antigen Processing Ubiquitination Proteasome Degradation
Modified AA Metabolism
Protein Mod. By Small Protein Removal
Mitotic Metaphase & Anaphase
Mitotic Spindle Org.
Alcohol Biosynthesis
Peptide Metabolism
M Phase
Autophagosome Org.
Nucleotide Excision Repair
Global Genome Nucleotide Excision Repair Gg Ner
Cardiac Muscle Cell Action Potential
Nuclear Transport
Cell Activation Involved In Immune Response
Resp. to Carbohydrate
Nucleocytoplasmic Transport
Pten Regulation
Nuclear Events Mediated By Nfe2l2
Fc Epsilon Receptor Fceri Signaling
Reg. Transmembrane Receptor Protein Serine Threonine Kinase Signaling
−1.5
−1.0
−0.5
0.0
signed −log10 p (positive: up)
GO_Slim
GOBP
Reactome
proteins
20
40
60
80
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

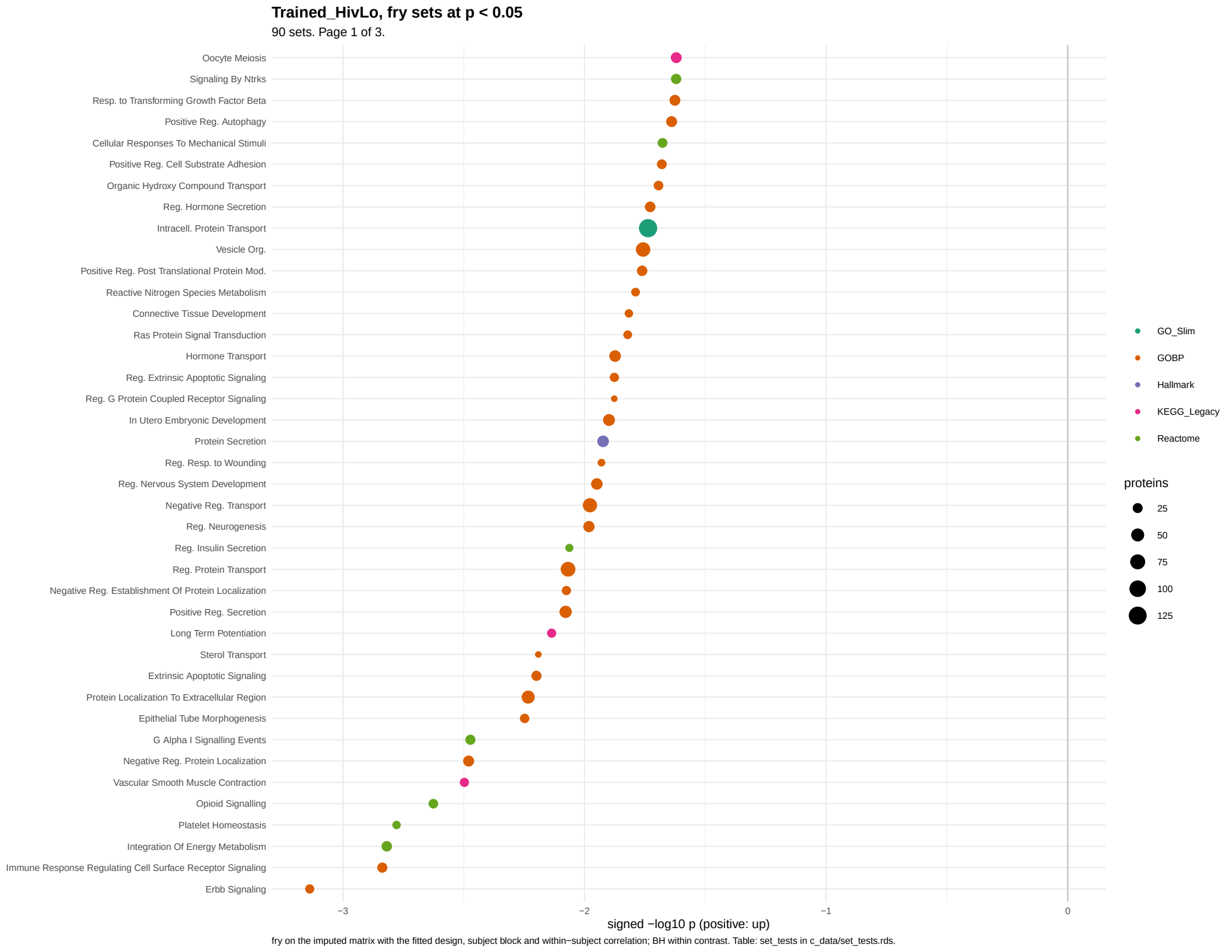

Baseline_HivLo, fry sets at p < 0.05
184 sets. Page 4 of 5.
Resp. ETC
B Cell Receptor Signaling
Hedgehog Off State
Protein Export From Nucleus
Reg. Ub−Dep. Protein Catabolism
Spop Mediated Proteas. Degradation Of PD−L1 CD274
Protein Folding
Transcriptional Regulation By Runx1
Abc Family Proteins Mediated Transport
mTOR Signalling
Positive Reg. Transmembrane Transport
Tcr Signaling
Vegf Signaling
Vacuole Org.
Xenobiotic Metabolism
Potential Therapeutics For Sars
Clathrin Endocytosis
Positive Reg. Hormone Secretion
Resp. to Temperature Stimulus
Calcium Mediated Signaling
Fceri Mediated Nf Kb Activation
Autophagy
Negative Reg. Monoatomic Ion Transport
S Phase
Interleukin 1 Family Signaling
Hormone Mediated Signaling
Cellular Oxidant Detoxification
Hiv Infection
Positive Reg. Ub−Dep. Protein Catabolism
Reg. Pten Stability & Activity
MT Cytoskel. Org. Involved In Mitosis
Circadian Clock
Reg. mRNA Stability By Proteins That Bind Au Rich Elements
Cellular Resp. to Steroid Hormone Stimulus
Reg. Intracell. Protein Transport
Reg. Exocytosis
Glycerophospholipid Biosynthesis
Vacuolar Transport
Cellular Resp. to Abiotic Stimulus
−1.5
−1.0
−0.5
0.0
0.5
1.0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
30
60
90
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

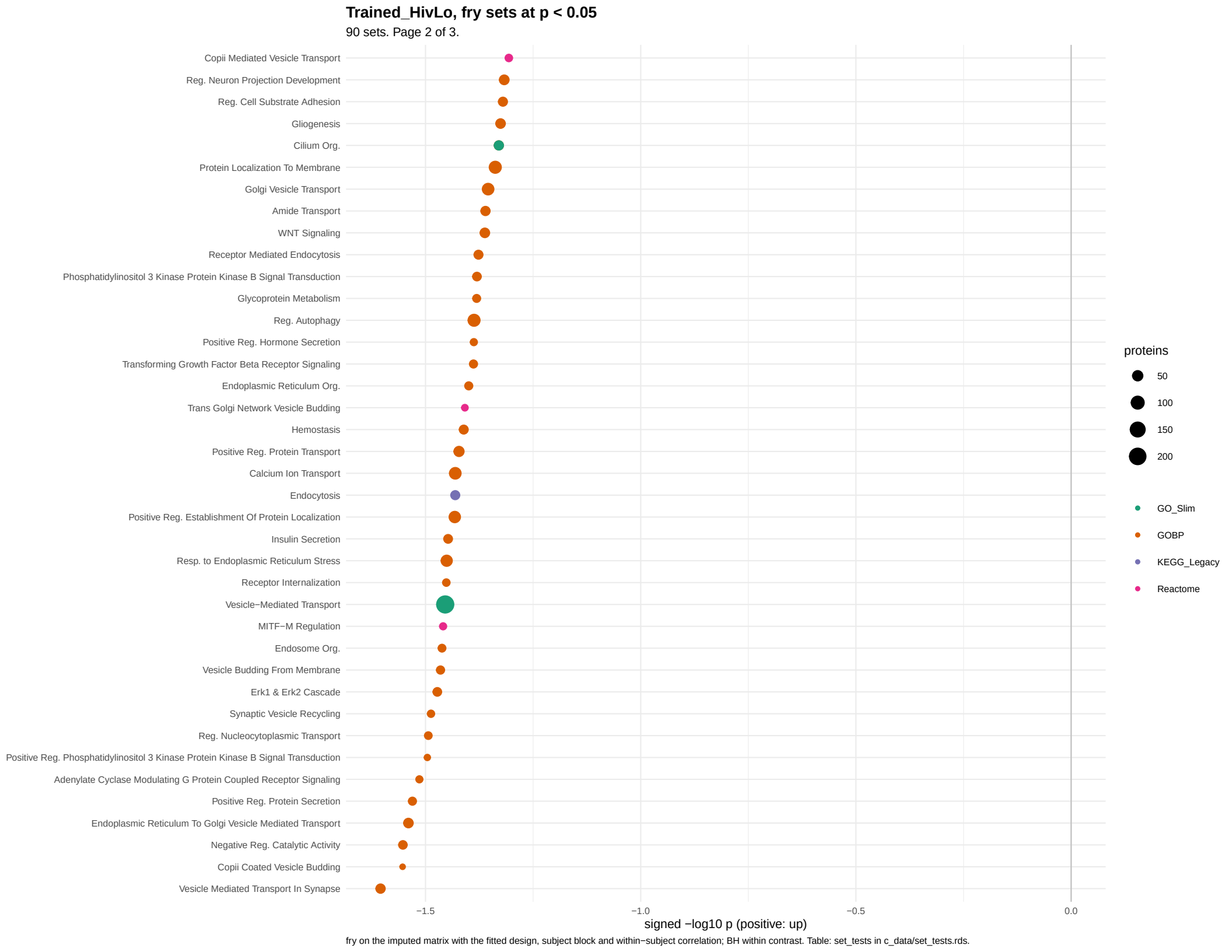
Baseline_HivLo, fry sets at p < 0.05

184 sets. Page 5 of 5.



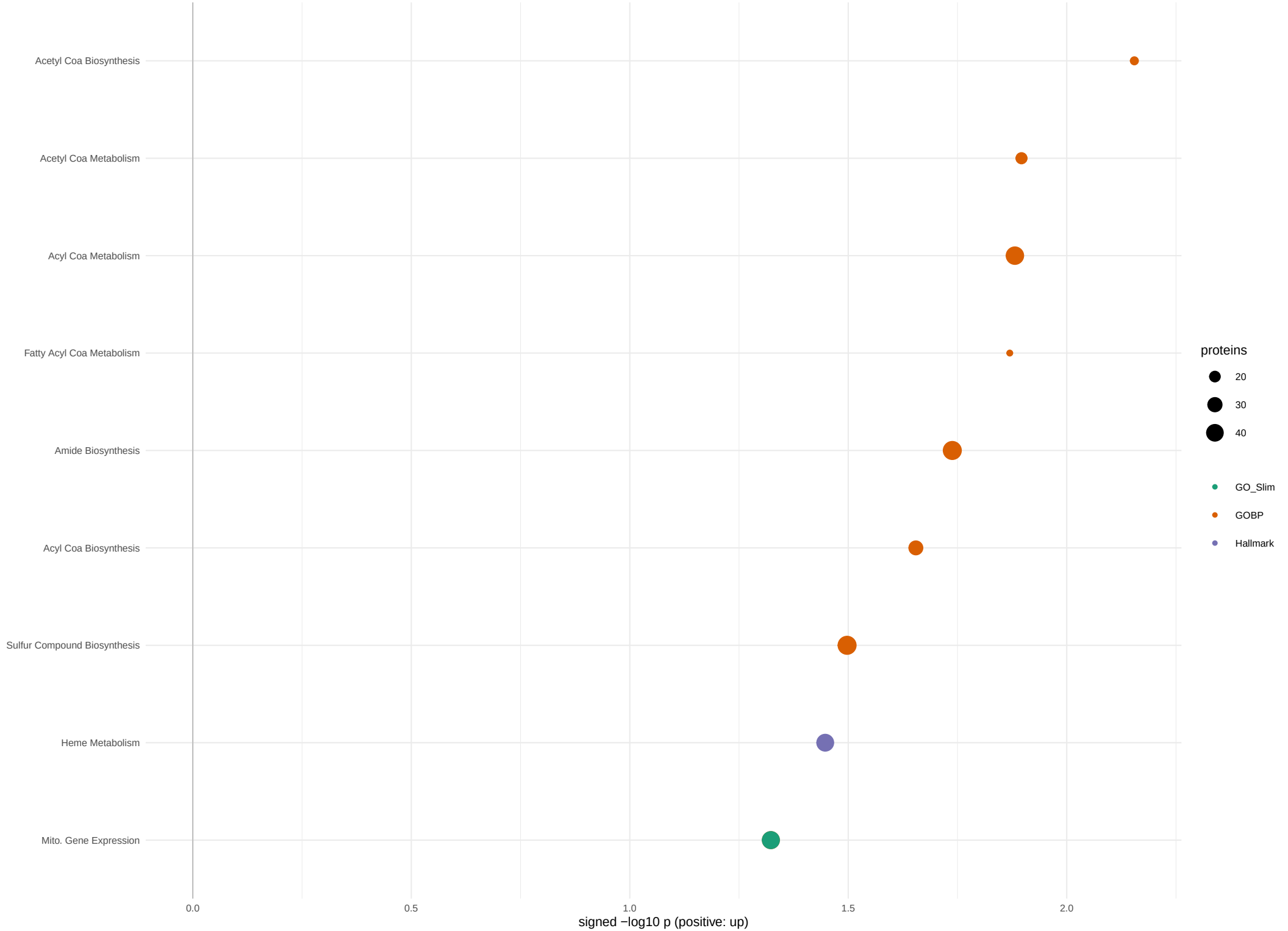
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Trained_HivLo, fry sets at p < 0.05
90 sets. Page 1 of 3.
Oocyte Meiosis
Signaling By Ntrks
Resp. to Transforming Growth Factor Beta
Positive Reg. Autophagy
Cellular Responses To Mechanical Stimuli
Positive Reg. Cell Substrate Adhesion
Organic Hydroxy Compound Transport
Reg. Hormone Secretion
Intracell. Protein Transport
Vesicle Org.
Positive Reg. Post Translational Protein Mod.
Reactive Nitrogen Species Metabolism
Connective Tissue Development
Ras Protein Signal Transduction
Hormone Transport
Reg. Extrinsic Apoptotic Signaling
Reg. G Protein Coupled Receptor Signaling
In Utero Embryonic Development
Protein Secretion
Reg. Resp. to Wounding
Reg. Nervous System Development
Negative Reg. Transport
Reg. Neurogenesis
Reg. Insulin Secretion
Reg. Protein Transport
Negative Reg. Establishment Of Protein Localization
Positive Reg. Secretion
Long Term Potentiation
Sterol Transport
Extrinsic Apoptotic Signaling
Protein Localization To Extracellular Region
Epithelial Tube Morphogenesis
G Alpha I Signalling Events
Negative Reg. Protein Localization
Vascular Smooth Muscle Contraction
Opioid Signalling
Platelet Homeostasis
Integration Of Energy Metabolism
Immune Response Regulating Cell Surface Receptor Signaling
Erbb Signaling
−3
−2
−1
0
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
25
50
75
100
125
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

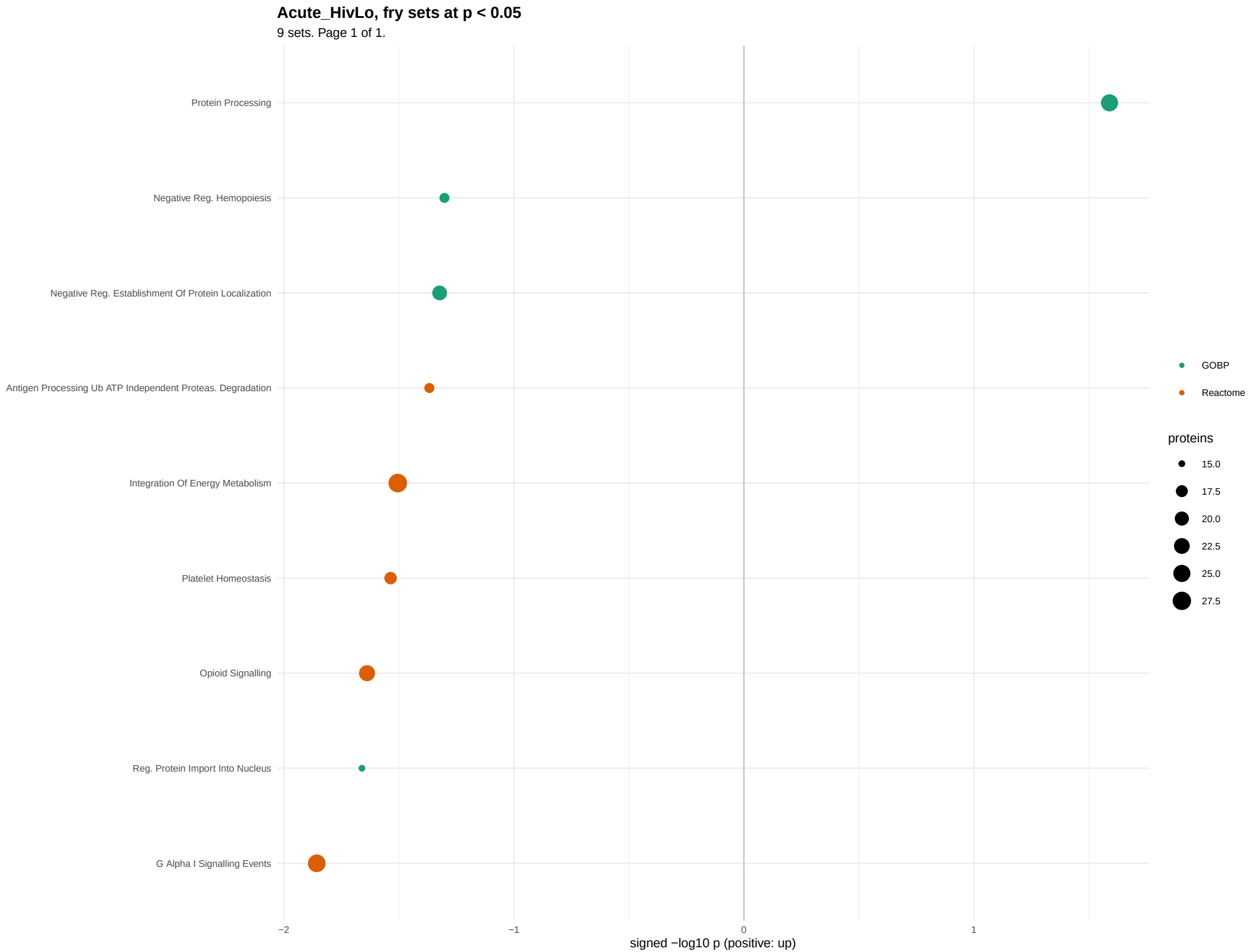

Trained_HivLo, fry sets at p < 0.05
90 sets. Page 2 of 3.
Copii Mediated Vesicle Transport
Reg. Neuron Projection Development
Reg. Cell Substrate Adhesion
Gliogenesis
Cilium Org.
Protein Localization To Membrane
Golgi Vesicle Transport
Amide Transport
WNT Signaling
Receptor Mediated Endocytosis
Phosphatidylinositol 3 Kinase Protein Kinase B Signal Transduction
Glycoprotein Metabolism
Reg. Autophagy
Positive Reg. Hormone Secretion
Transforming Growth Factor Beta Receptor Signaling
Endoplasmic Reticulum Org.
Trans Golgi Network Vesicle Budding
Hemostasis
Positive Reg. Protein Transport
Calcium Ion Transport
Endocytosis
Positive Reg. Establishment Of Protein Localization
Insulin Secretion
Resp. to Endoplasmic Reticulum Stress
Receptor Internalization
Vesicle−Mediated Transport
MITF−M Regulation
Endosome Org.
Vesicle Budding From Membrane
Erk1 & Erk2 Cascade
Synaptic Vesicle Recycling
Reg. Nucleocytoplasmic Transport
Positive Reg. Phosphatidylinositol 3 Kinase Protein Kinase B Signal Transduction
Adenylate Cyclase Modulating G Protein Coupled Receptor Signaling
Positive Reg. Protein Secretion
Endoplasmic Reticulum To Golgi Vesicle Mediated Transport
Negative Reg. Catalytic Activity
Copii Coated Vesicle Budding
Vesicle Mediated Transport In Synapse
−1.5
−1.0
−0.5
0.0
signed −log10 p (positive: up)
proteins
50
100
150
200
GO_Slim
GOBP
KEGG_Legacy
Reactome
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

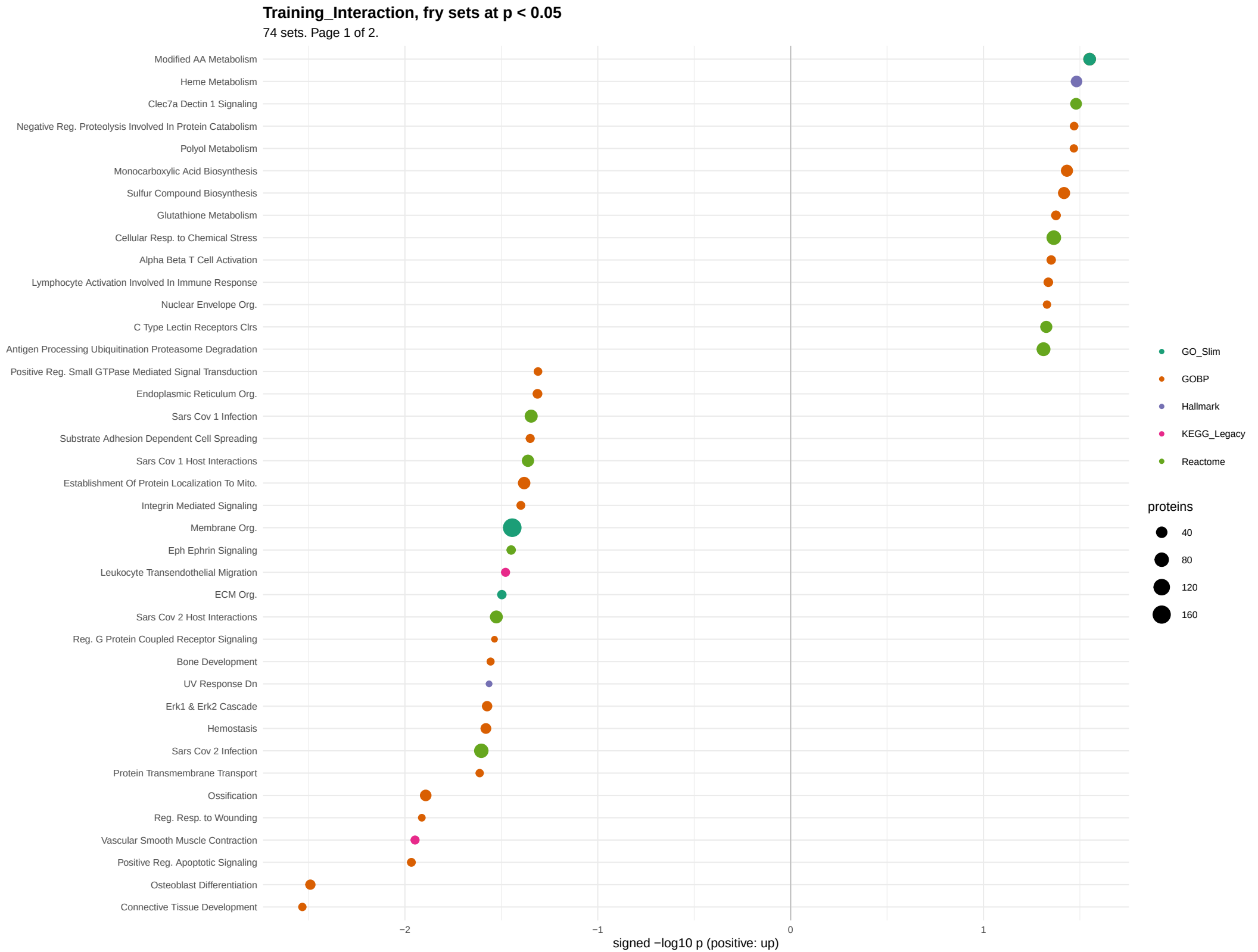
Trained_HivLo, fry sets at p < 0.05

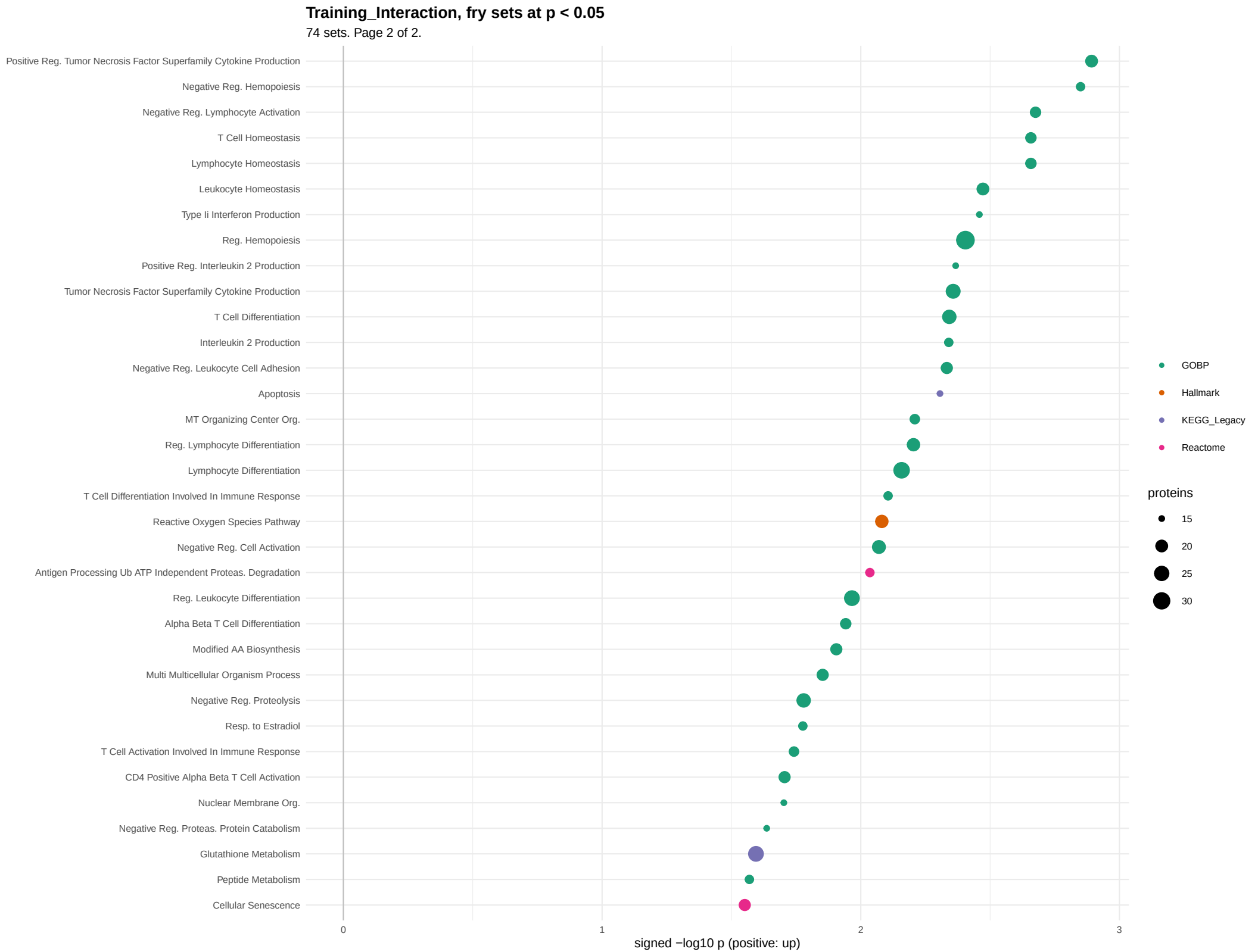
90 sets. Page 3 of 3.

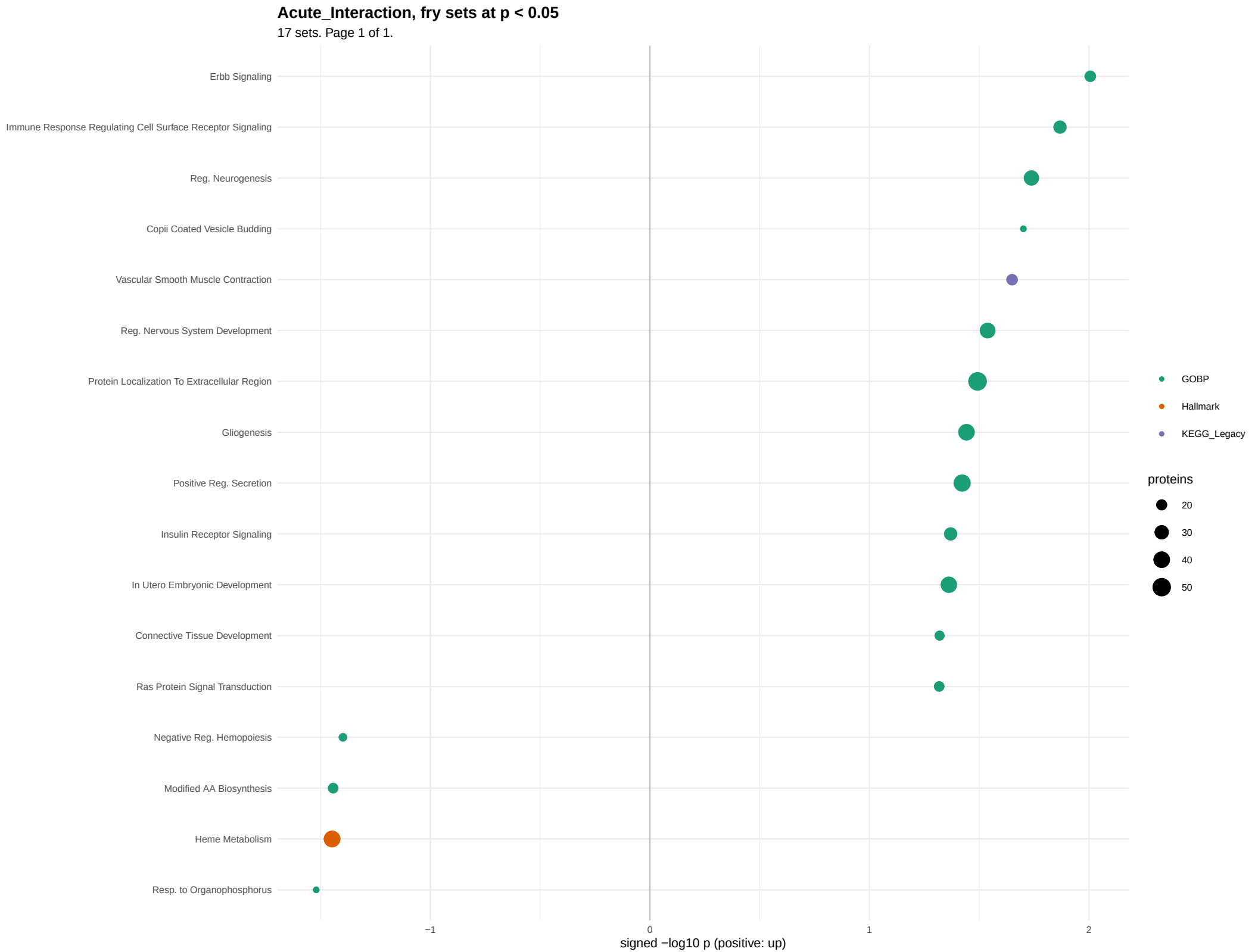


fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_HivLo, fry sets at p < 0.05
9 sets. Page 1 of 1.
Protein Processing
Negative Reg. Hemopoiesis
Negative Reg. Establishment Of Protein Localization
Antigen Processing Ub ATP Independent Proteas. Degradation
Integration Of Energy Metabolism
Platelet Homeostasis
Opioid Signalling
Reg. Protein Import Into Nucleus
G Alpha I Signalling Events
−2
−1
0
1
signed −log10 p (positive: up)
GOBP
Reactome
proteins
15.0
17.5
20.0
22.5
25.0
27.5
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

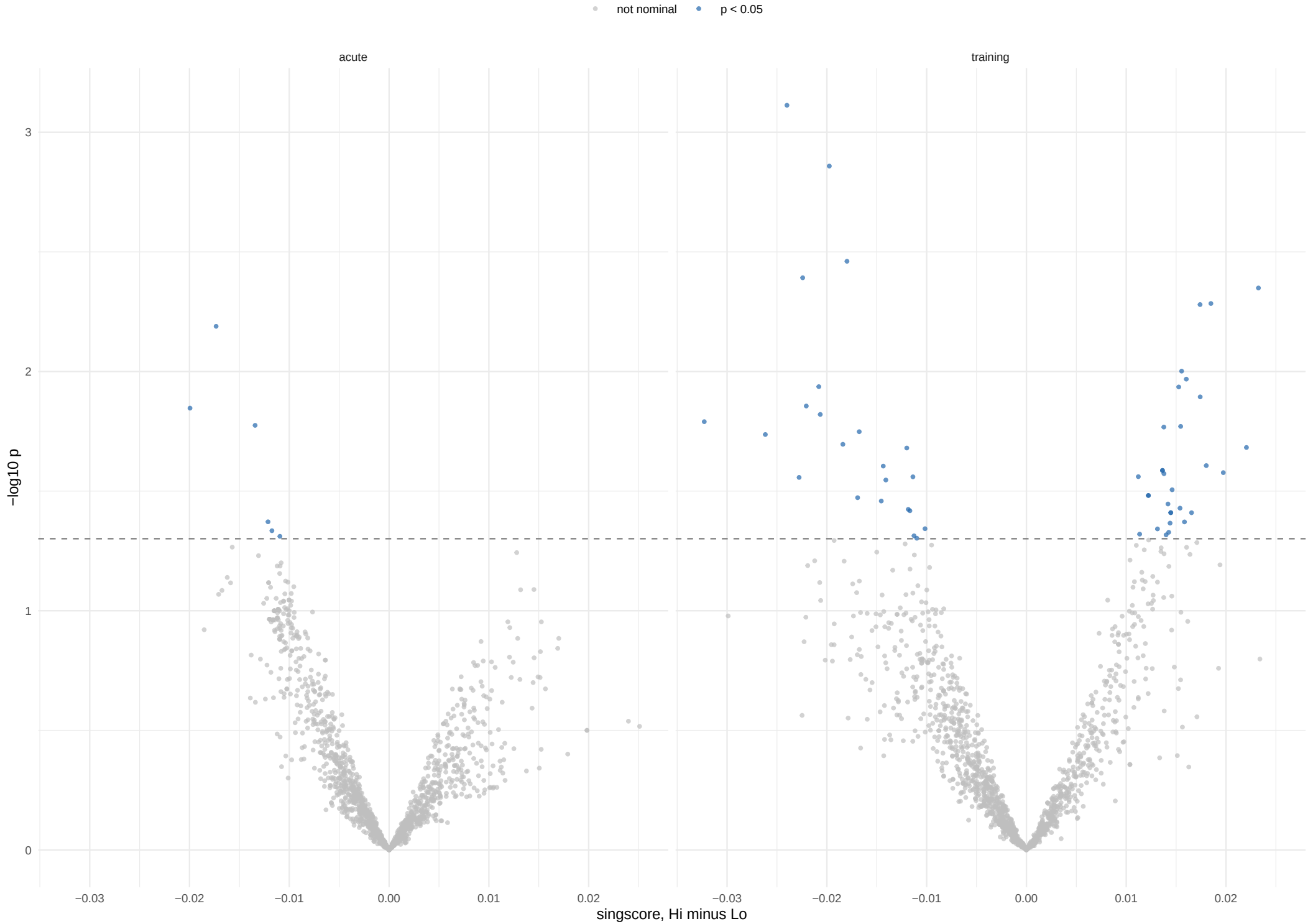

Training_Interaction, fry sets at p < 0.05
74 sets. Page 1 of 2.
Modified AA Metabolism
Heme Metabolism
Clec7a Dectin 1 Signaling
Negative Reg. Proteolysis Involved In Protein Catabolism
Polyol Metabolism
Monocarboxylic Acid Biosynthesis
Sulfur Compound Biosynthesis
Glutathione Metabolism
Cellular Resp. to Chemical Stress
Alpha Beta T Cell Activation
Lymphocyte Activation Involved In Immune Response
Nuclear Envelope Org.
C Type Lectin Receptors Clrs
Antigen Processing Ubiquitination Proteasome Degradation
Positive Reg. Small GTPase Mediated Signal Transduction
Endoplasmic Reticulum Org.
Sars Cov 1 Infection
Substrate Adhesion Dependent Cell Spreading
Sars Cov 1 Host Interactions
Establishment Of Protein Localization To Mito.
Integrin Mediated Signaling
Membrane Org.
Eph Ephrin Signaling
Leukocyte Transendothelial Migration
ECM Org.
Sars Cov 2 Host Interactions
Reg. G Protein Coupled Receptor Signaling
Bone Development
UV Response Dn
Erk1 & Erk2 Cascade
Hemostasis
Sars Cov 2 Infection
Protein Transmembrane Transport
Ossification
Reg. Resp. to Wounding
Vascular Smooth Muscle Contraction
Positive Reg. Apoptotic Signaling
Osteoblast Differentiation
Connective Tissue Development
−2
−1
0
1
signed −log10 p (positive: up)
GO_Slim
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
40
80
120
160
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Training_Interaction, fry sets at p < 0.05
74 sets. Page 2 of 2.
Positive Reg. Tumor Necrosis Factor Superfamily Cytokine Production
Negative Reg. Hemopoiesis
Negative Reg. Lymphocyte Activation
T Cell Homeostasis
Lymphocyte Homeostasis
Leukocyte Homeostasis
Type Ii Interferon Production
Reg. Hemopoiesis
Positive Reg. Interleukin 2 Production
Tumor Necrosis Factor Superfamily Cytokine Production
T Cell Differentiation
Interleukin 2 Production
Negative Reg. Leukocyte Cell Adhesion
Apoptosis
MT Organizing Center Org.
Reg. Lymphocyte Differentiation
Lymphocyte Differentiation
T Cell Differentiation Involved In Immune Response
Reactive Oxygen Species Pathway
Negative Reg. Cell Activation
Antigen Processing Ub ATP Independent Proteas. Degradation
Reg. Leukocyte Differentiation
Alpha Beta T Cell Differentiation
Modified AA Biosynthesis
Multi Multicellular Organism Process
Negative Reg. Proteolysis
Resp. to Estradiol
T Cell Activation Involved In Immune Response
CD4 Positive Alpha Beta T Cell Activation
Nuclear Membrane Org.
Negative Reg. Proteas. Protein Catabolism
Glutathione Metabolism
Peptide Metabolism
Cellular Senescence
0
1
2
3
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
Reactome
proteins
15
20
25
30
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.


Acute_Interaction, fry sets at p < 0.05
17 sets. Page 1 of 1.
Erbb Signaling
Immune Response Regulating Cell Surface Receptor Signaling
Reg. Neurogenesis
Copii Coated Vesicle Budding
Vascular Smooth Muscle Contraction
Reg. Nervous System Development
Protein Localization To Extracellular Region
Gliogenesis
Positive Reg. Secretion
Insulin Receptor Signaling
In Utero Embryonic Development
Connective Tissue Development
Ras Protein Signal Transduction
Negative Reg. Hemopoiesis
Modified AA Biosynthesis
Heme Metabolism
Resp. to Organophosphorus
−1
0
1
2
signed −log10 p (positive: up)
GOBP
Hallmark
KEGG_Legacy
proteins
20
30
40
50
fry on the imputed matrix with the fitted design, subject block and within−subject correlation; BH within contrast. Table: set_tests in c_data/set_tests.rds.

ANCOVA: Hi minus Lo at the later time point, at equal starting value

Training: 53 of 1378 sets at p < 0.05. Acute: 6 of 1378.



lm(later ~ earlier + arm) per set; training is T2 on T1, acute T3 on T2. Dashed line: p = 0.05. BH within window and collection. Table: c_data/03_contrasts.xlsx, ancova.